Predicting Confidence in Social Institutions: A Classification Analysis Using World Values Survey Data

Generative AI was used in this assignment
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1. Data Exploration:

Figure 1: Overall Distribution of Predictor Variables

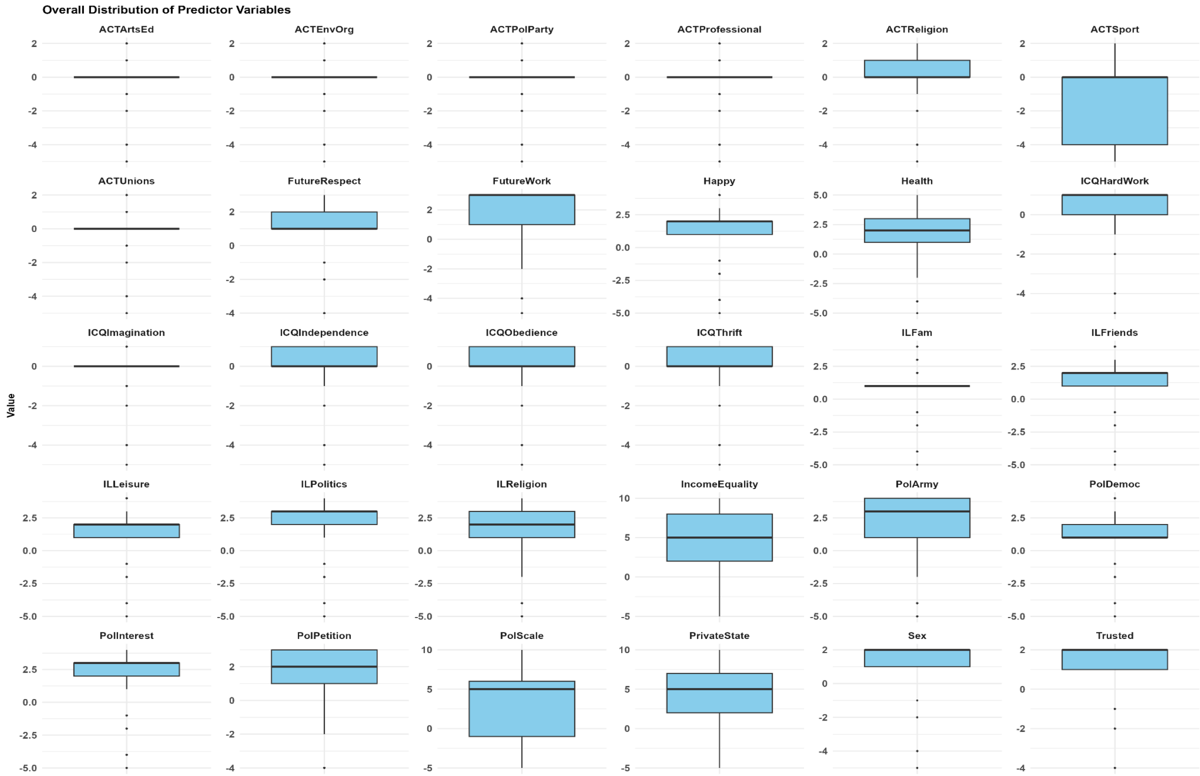


Figure 1 reveals that predictors vary considerably in their spread and behaviour. PolScale, PrivateState, and IncomeEquality span a wide range of approximately −5 to 10, suggesting these variables capture meaningful differences across respondents and may be more useful for separating confidence groups. Activity-related variables including ACTArtsEd, ACTEnvOrg, ACTPolParty, ACTProfessional, ACTUnions, and ICQImagination are heavily concentrated near zero, meaning most respondents gave very similar responses. This limited variation may reduce their ability to distinguish between Low and High confidence classes. Variables such as Happy, Health, ILFriends, ILLeisure, and PolInterest show moderate spread that could still contribute meaningfully to the models.

Figure 2: Distribution of High and Low Confidence Classes Across Three Binary Class Variables

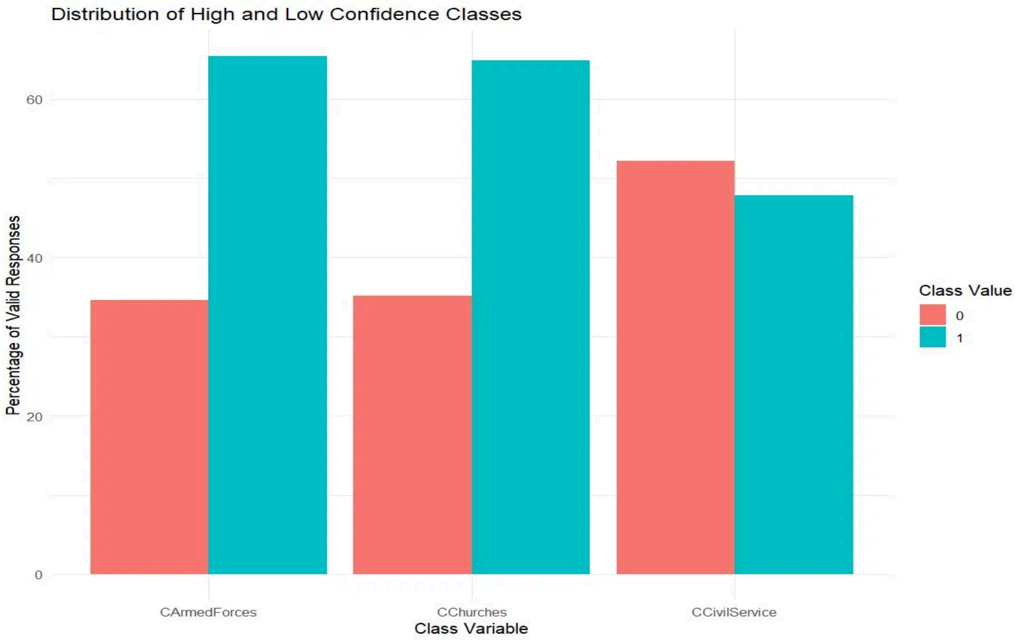


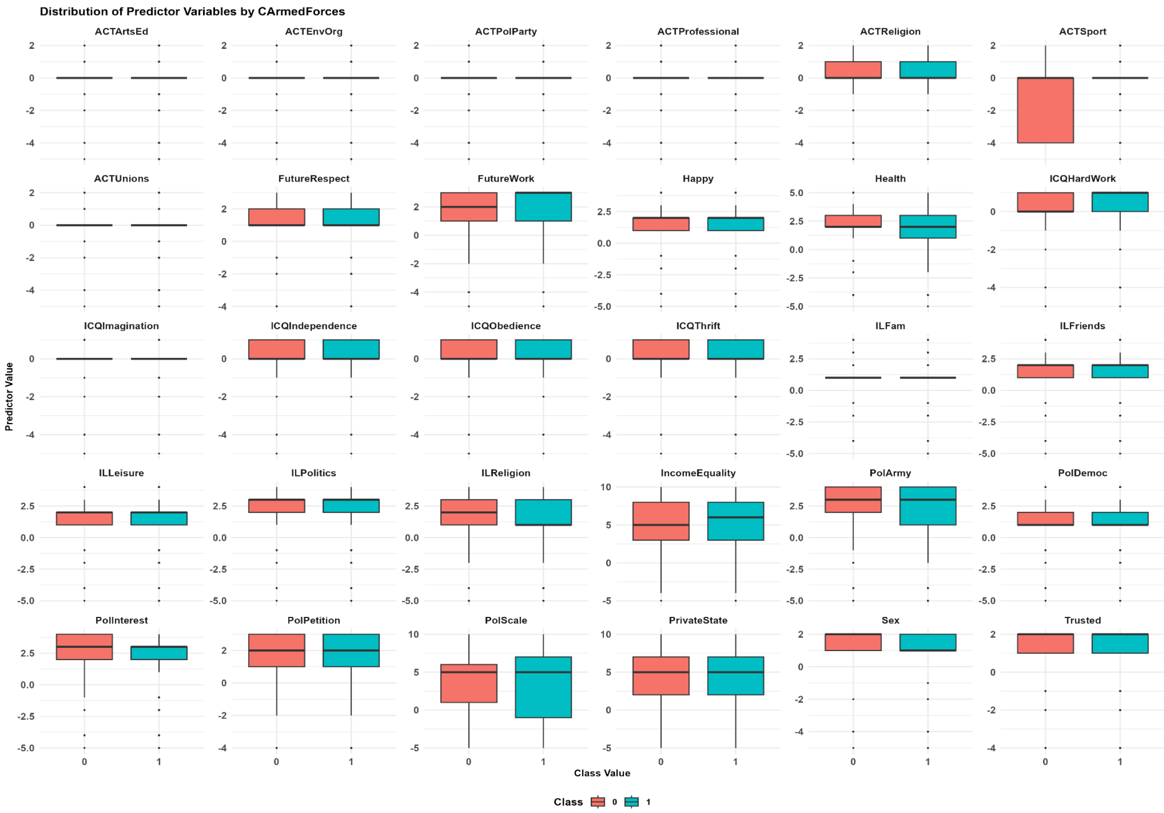
Table 1: Distribution of High and Low Confidence Classes

Class Variable	Low Confidence, 0	High Confidence, 1
CCivilService	52.19%	47.81%
CChurches	35.12%	64.88%
CArmedForces	34.60%	65.40%

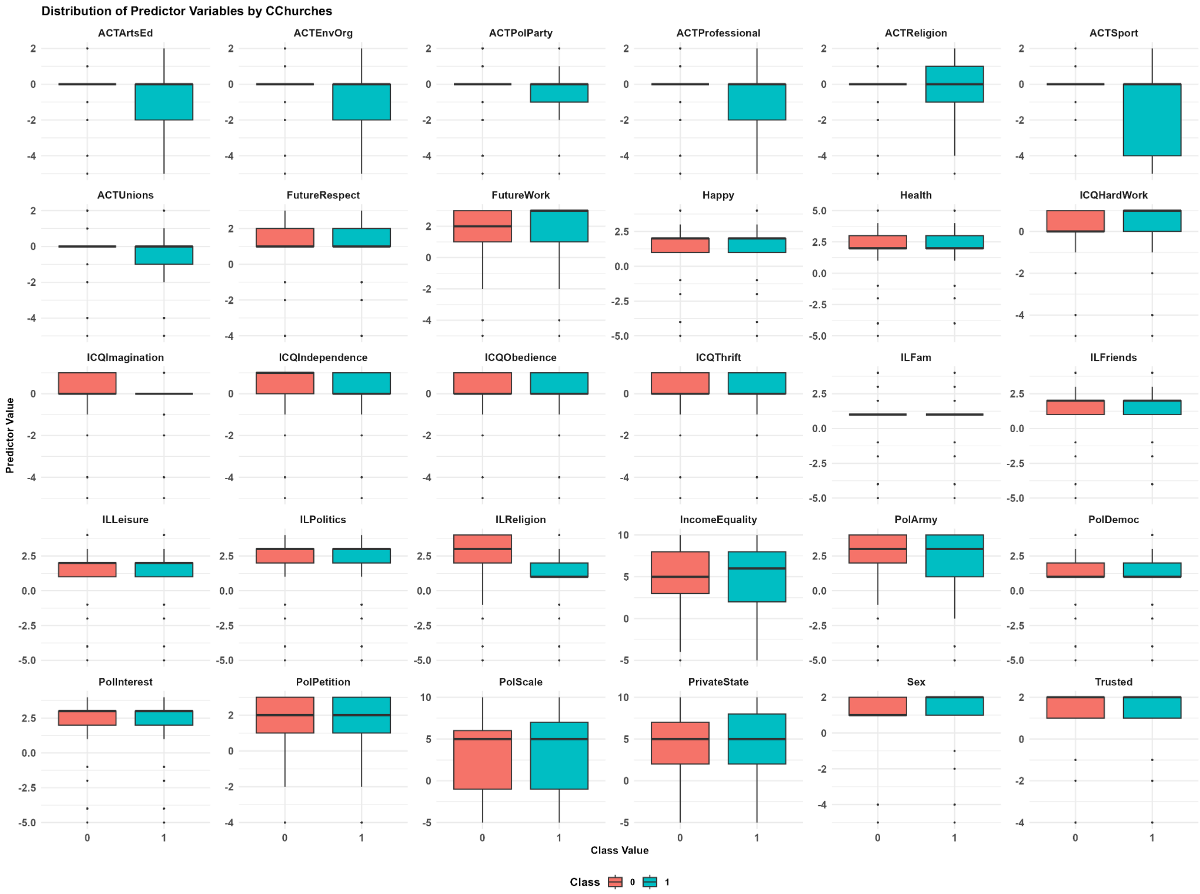
As shown in Figure 2 and Table 1, CCivilService is the most balanced class variable (Low: 52.19%, High: 47.81%), meaning models predicting it are less likely to be biased toward one class. CChurches and CArmedForces both show moderate imbalance, with high-confidence responses making up 64.88% and 65.40% of valid responses respectively, suggesting respondents generally tend to place more trust in churches and the armed forces than in the civil service. Because of this imbalance, accuracy alone will be misleading, and F1-score, precision, and recall will be more appropriate evaluation metrics.

Figure 3 Distribution of Predictor Variables by Classes

a) CArmedForces



b) CChurches


Distribution of Predictor Variables by CChurches
ACTArtsEd
ACTEnvOrg
ACTPolParty
ACTProfessional
ACTReligion
ACTSport
ACTUnions
FutureRespect
FutureWork
Happy
Health
ICQHardWork
ICQImagination
ICQIndependence
ICQObedience
ICQThrift
ILFam
ILFriends
ILLeisure
ILPolitics
ILReligion
IncomeEquality
PolArmy
PolDemoc
PolInterest
PolPetition
PolScale
PrivateState
Sex
Trusted
Predictor Value
Class Value
Class 0 1

c) CCivilService

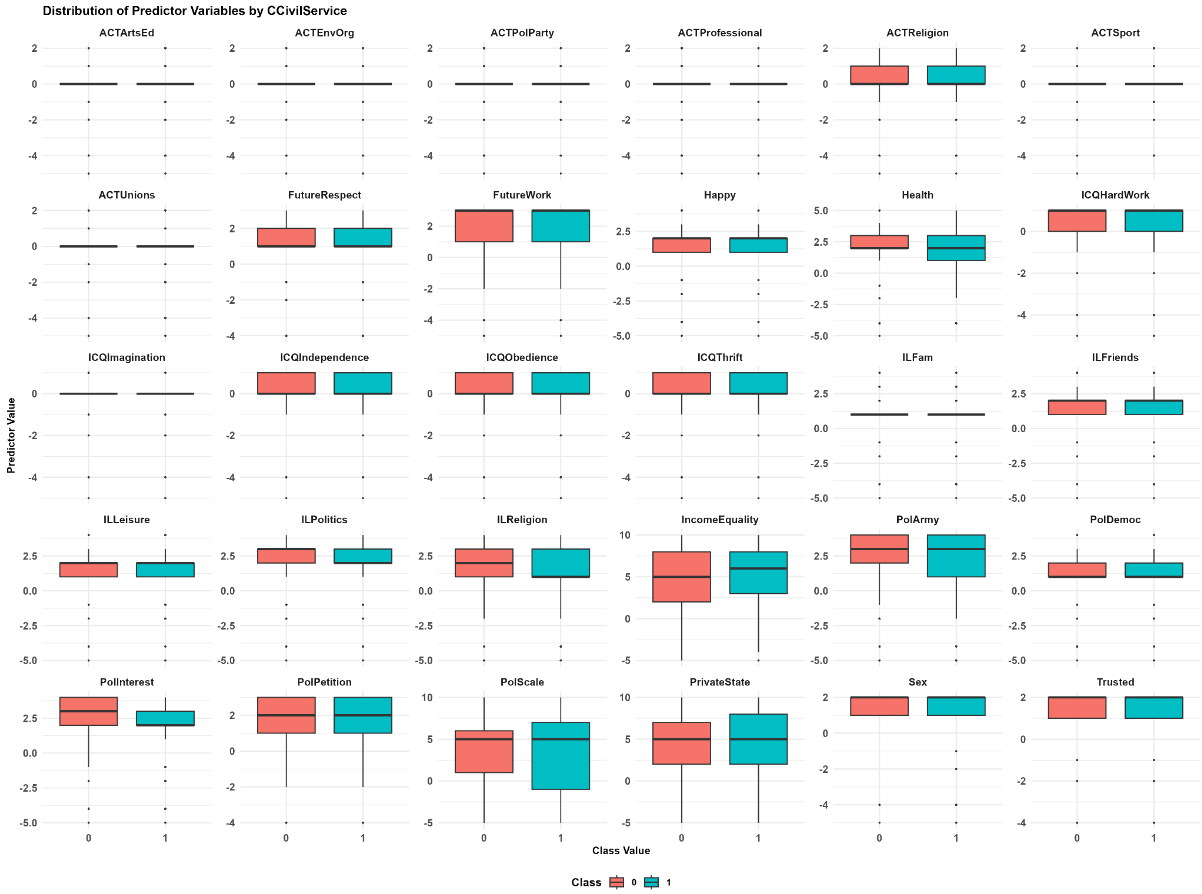


Figure 3 compares predictor distributions across Low and High confidence groups for each class variable. The clearest separation appears in Figure 3b) CChurches, where ILReligion and ACTReligion show noticeably different distributions between the two groups. Respondents who place greater importance on religion and are more religiously active tend to report higher confidence in churches, which is an intuitive finding.Figure 3a) for CArmedForces, political variables such as PolArmy, PolInterest, and PolScale show some separation, suggesting that political attitudes and engagement are associated with confidence in the armed forces.Figure 3c) for CCivilService shows the least separation overall, with only IncomeEquality, PrivateState, and Health showing modest differences, indicating that this class variable may be inherently harder to predict from the available attributes.

Figure 4: Correlation Matrix of Predictor Variables

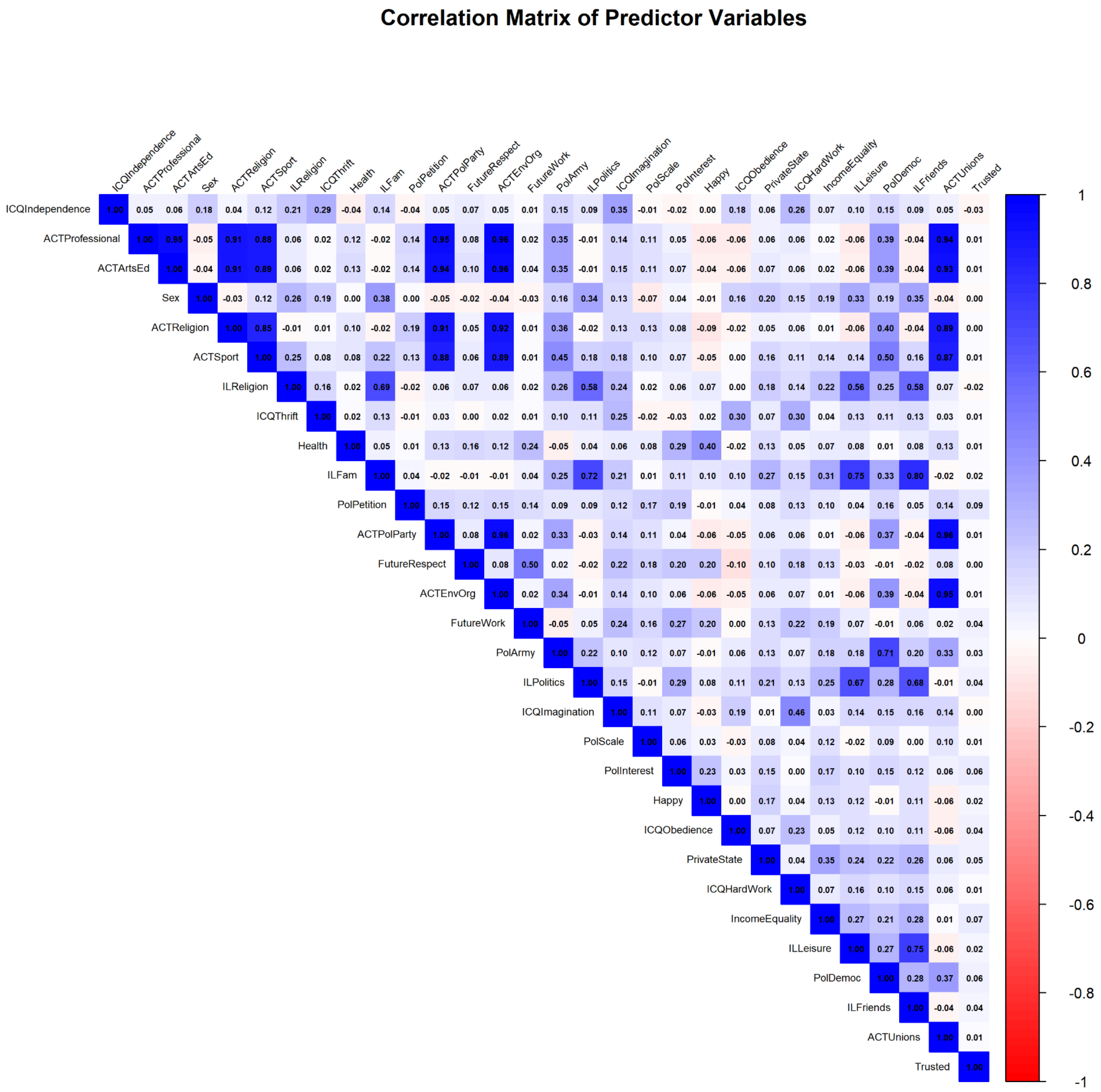
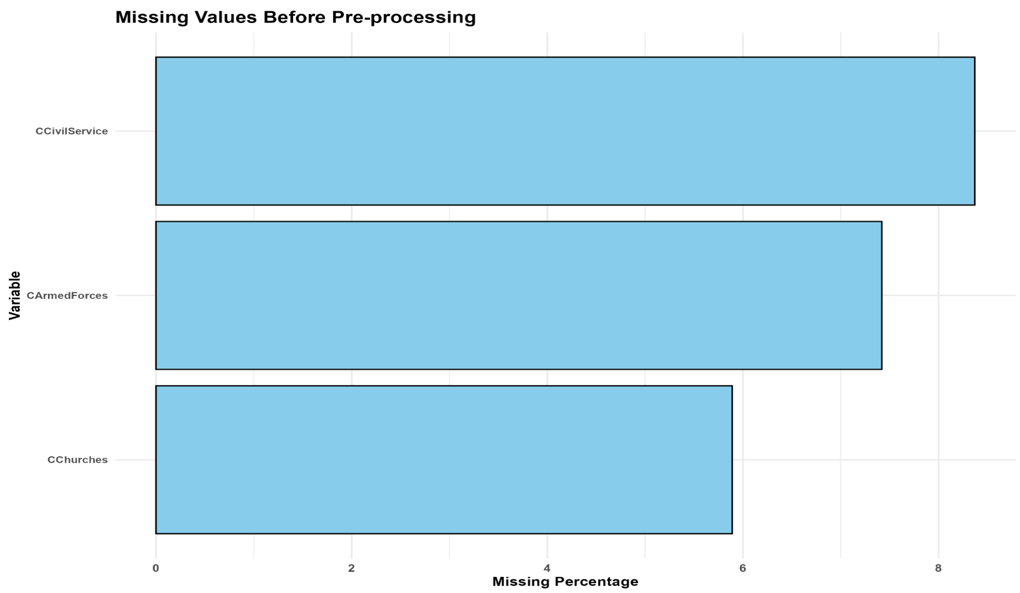


Figure 4 highlights strong redundancy among the activity-related predictors. ACTPolParty and ACTEnvOrg are correlated at r = 0.96, ACTEnvOrg and ACTUnions at r = 0.95, and ACTReligion and ACTEnvOrg at r = 0.92, suggesting these variables largely capture the same underlying pattern of organisational participation. Keeping all of them may not add much new information and could increase model complexity unnecessarily. Beyond the activity variables, PolArmy and PolDemoc (r = 0.71) and ILReligion and ILFam (r = 0.69) also show notable associations, reflecting relationships between political attitudes and between religious and family values respectively.

Overall, the exploratory analysis identified several important considerations for model training. CChurches and CArmedForces exhibit moderate class imbalance, with high-confidence responses comprising approximately two-thirds of valid responses for each variable, meaning evaluation metrics beyond accuracy will be necessary. Several activity-related predictors show limited variation, and strong inter-predictor correlations among the ACT variables suggest redundancy that should be reviewed during feature selection. Country and Wave were excluded as predictors, as including them could allow models to exploit national and time-period effects rather than the individual-level attitudinal patterns that are the focus of this analysis.

2. Data Pre - Processing:

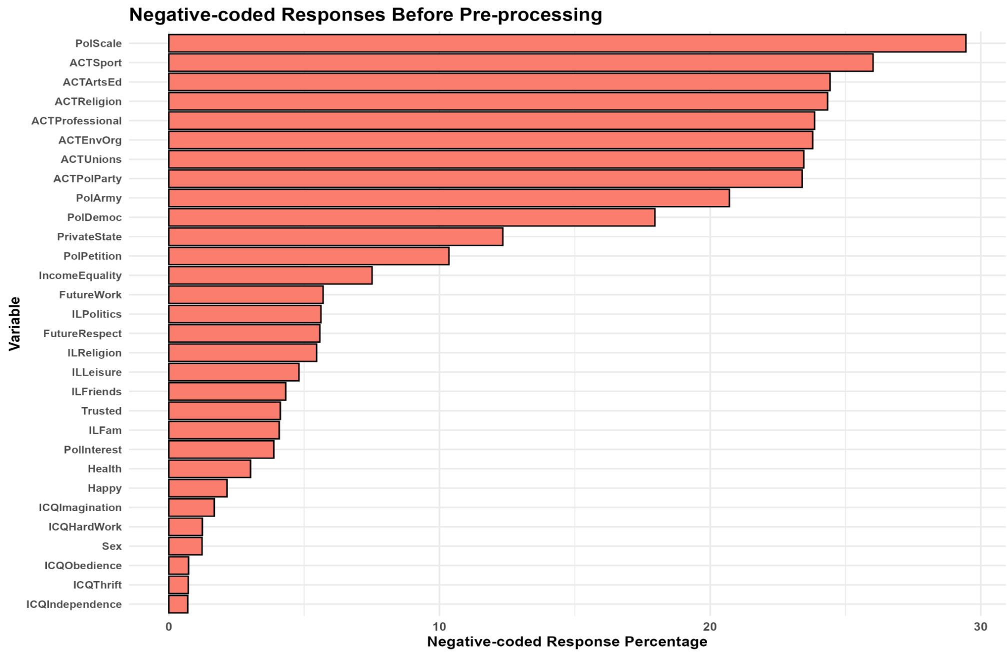
Figure 5: Missing Values Before Pre-processing



Several pre-processing steps were carried out to prepare the dataset for model fitting.

First, missing values were checked before any recoding. As shown in Figure 5, missing values were present only in the three class variables at this stage, with CCivilService having the highest missing percentage (8.37%), followed by CArmedForces (7.42%) and CChurches (5.89%).

Figure 6: Negative-coded Responses Before Pre-processing



Second, negative-coded WVS responses were identified across the predictor variables. As shown in Figure 6, these do not represent valid responses but rather non-substantive survey codes such as not asked or no answer. PolScale had the highest proportion at approximately 30%, followed by several activity-related variables including ACTSport, ACTArtsEd, ACTReligion, ACTProfessional, ACTEnvOrg, ACTUnions, and ACTPolParty, all exceeding 20%. These values were recoded to NA to prevent them from being treated as meaningful categories during model training.

Third, complete-case filtering was applied, retaining only observations with no missing values across all predictors and class variables. This reduced the dataset from 20,000 to 7,952 observations.

Fourth, all class variables and predictor variables were converted to factors, as the WVS responses are categorical or ordinal rather than continuous.

Figure 7: Class Balance After Pre-processing

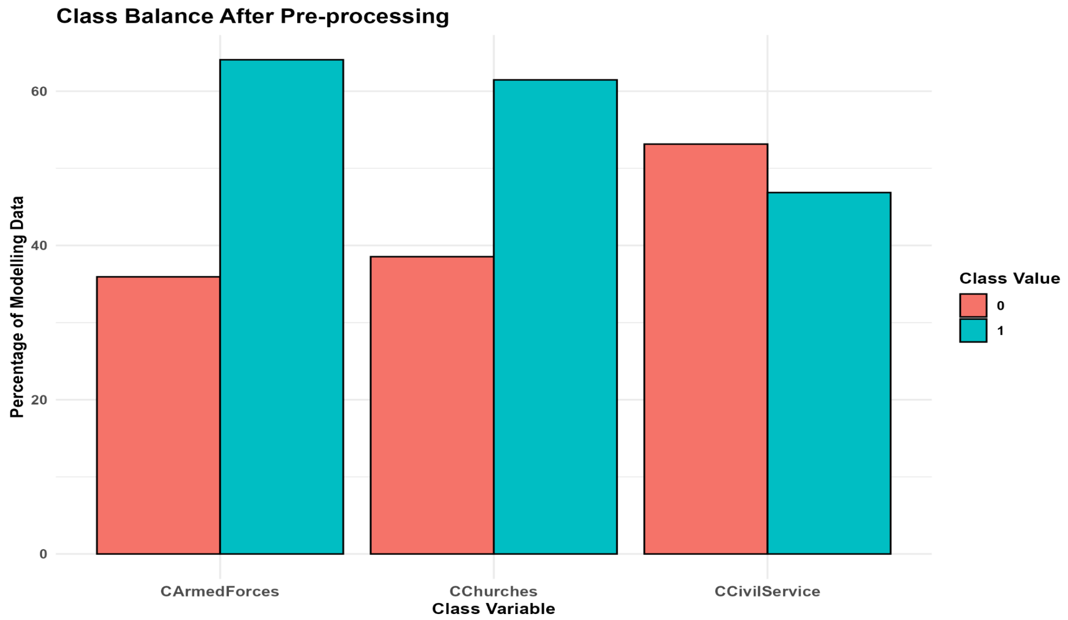


Table 2: Class Balance After Pre-processing

Class Variable	Low Confidence, 0	High Confidence, 1
CCivilService	53.14%	46.86%
CChurches	38.54%	61.46%
CArmedForces	35.93%	64.07%

Finally, as shown in Table 2 and Figure 7, class distributions only slightly changed after pre-processing. CCivilService remained somewhat balanced (Low: 53.14%, High: 46.86%), while CChurches (Low: 38.54%, High: 61.46%) and CArmedForces (Low: 35.93%, High: 64.07%) retained their moderate imbalance, confirming that complete-case filtering did not introduce any meaningful bias into the class proportions.

3. Data Splitting:

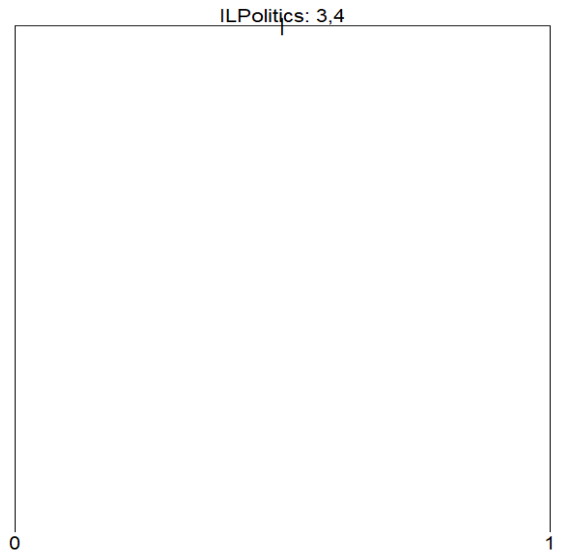
The cleaned dataset was split into training and testing subsets using a 70:30 split with set.seed(34091904) to ensure reproducibility. The training set contained 5,566 observations and was used to fit the classification models, while the test set contained 2,386 observations and was used to evaluate model performance. This separation helps assess how well the models generalise to unseen data.

4.Classification Model Implementation:

Decision Tree:

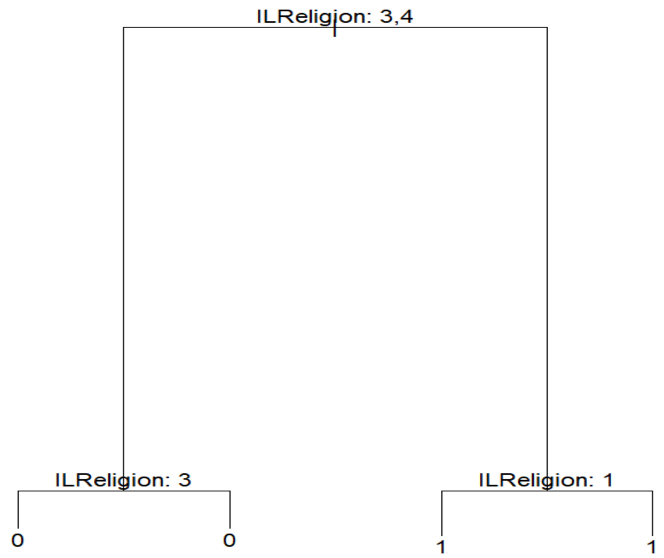
Decision tree models were fitted separately for each of the three class variables using the rpart package , with the remaining two class variables, Country, and Wave excluded from the predictor set in each case. The decision tree model was used as a single-tree classifier, where observations were split into groups based on predictor values, and final class predictions were made according to the terminal node classifications.

Figure 8: Decision Tree for Predicting Confidence in Civil Service



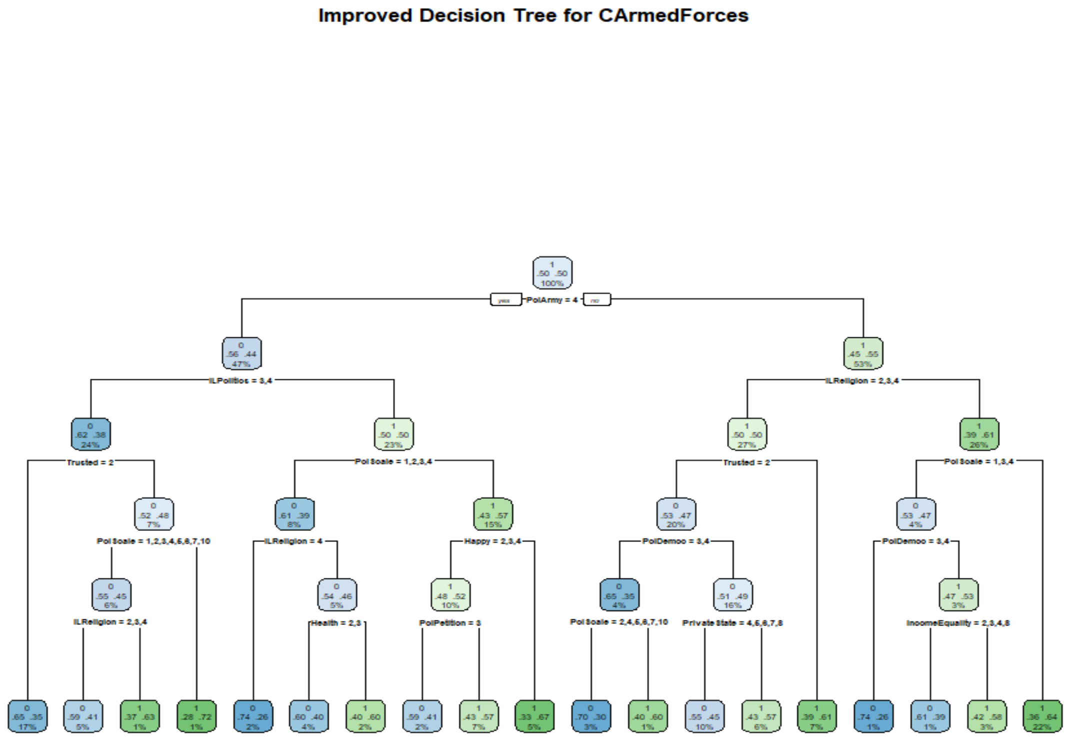
For CCivilService, the default tree produced a simple two-node structure using ILPolitics as the sole splitting variable, classifying respondents with ILPolitics categories 3 and 4 as High confidence and the remainder as Low confidence, as shown in Figure 8.

Figure 9: Decision Tree for Predicting Confidence in Churches



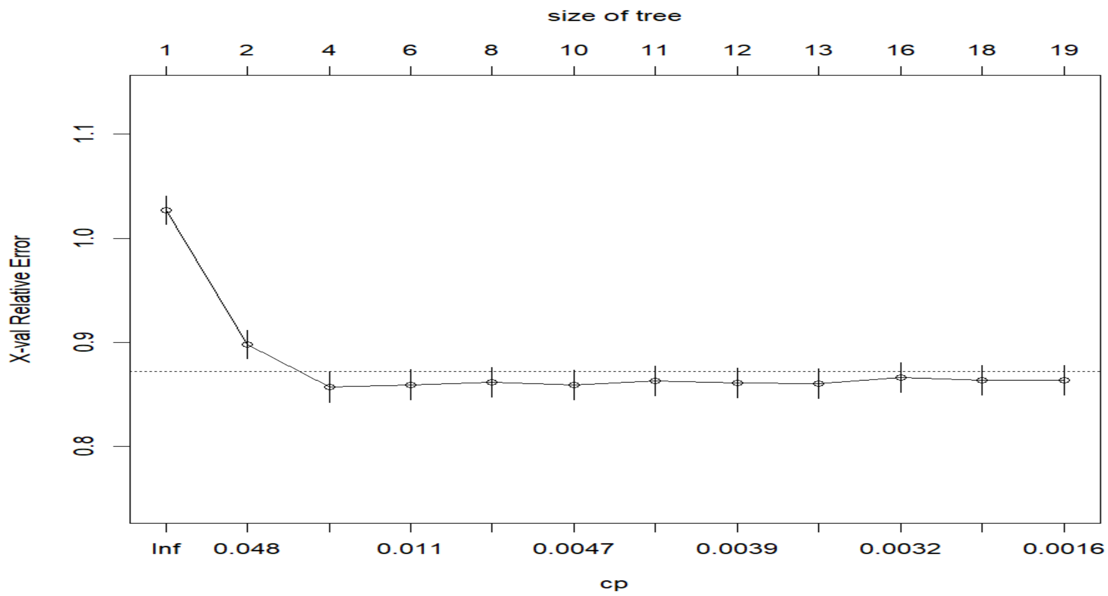
For CChurches, the tree used ILReligion as the primary splitting variable with further splits on ILReligion subcategories, as shown in Figure 9, which is intuitive given the expected relationship between religious importance and confidence in churches.

Figure 10: Improved Decision Tree for Predicting Confidence in Armed Forces



For CArmedForces, the default tree produced a single-node structure with no useful splits, so an improved model was fitted using rpart. Equal class priors (prior = c(0.5, 0.5)) were applied to prevent the model from defaulting to the majority class. cp = 0.001 allowed the tree to grow meaningfully, while minsplit = 40, minbucket = 15, and maxdepth = 5 controlled complexity and reduced the risk of overfitting. Ten-fold cross-validation was also applied to assess tree performance. The resulting tree, shown in Figure 10, identified multiple splitting variables including PolArmy, ILPolitics, ILReligion, Trusted, PolScale, PolDemoc, PrivateState, IncomeEquality, Health, and Happy. This suggests that confidence in the armed forces was not captured by one simple predictor, rather by multiple predictors.

Figure 11: Cross-validation Plot for the Improved Armed Forces Decision Tree



The cross-validation plot in Figure 11 shows that error decreased sharply beyond a single node before stabilising, confirming that a more developed tree structure was appropriate without risking excessive overfitting.

Naïve Bayes:

Table 3: Prior Class Probabilities from Naïve Bayes Models

Class Variable	Prior Probability for Class 0	Prior Probability for Class 1	Majority Class
CCivilService	0.5237	0.4763	Class 0
CChurches	0.382	0.618	Class 1
CArmedForces	0.3622	0.6378	Class 1

Naïve Bayes models were fitted separately for each class variable using the naiveBayes() function from the e1071 package with default settings and no tuning parameters adjusted.

The model estimates prior class probabilities and conditional probabilities for each predictor category, assigning observations to the class with the highest posterior probability. It assumes conditional independence between predictors, which may not fully hold given the correlations observed earlier, but Naïve Bayes remains a useful baseline classifier. Since all predictors were converted to factors during pre-processing, they were treated as discrete categorical variables.

As shown in Table 3, prior probabilities reflect the training data class distributions. CCivilService was the most balanced (Class 0: 0.5237, Class 1: 0.4763), while CChurches (0.6180) and CArmedForces (0.6378) both had class 1 as the majority.

Table 4: Implementation of Ensemble Models

Table 4 summarises the three ensemble methods used in the analysis: Bagging, Boosting and Random Forest. It explains how each method builds and combines multiple decision trees, and lists the key parameters and R packages used to implement each model.

Ensemble Method	**How It Works**	**Key Parameters**
Bagging	Builds multiple decision trees from bootstrap samples of the training data. Each bootstrap sample is drawn with replacement, so every tree is trained on a slightly different version of the training set. Final class prediction is made using majority voting across all trees.	mfinal = 100 (100 bootstrap decision trees built for each class variable) adabag package was used
Boosting	Builds trees sequentially, where each new tree gives more weight to observations that were misclassified by earlier trees. Final prediction is made using a weighted vote across all trees. More adaptive than bagging as it focuses more on difficult-to-classify observations during later iterations.	mfinal = 100 (100 boosted trees trained for each class variable) adabag package was used
Random Forest	Builds many decision trees and combines their predictions using majority voting. Unlike bagging, it also randomly selects a subset of predictors at each split, which reduces correlation between trees and improves model stability.	ntree = 500 (500 trees fitted per class variable); mtry = 5 (5 predictor variables randomly considered at each split) randomforest package was used

5. Model Performance Evaluation and Comparison

Figure 12: Confusion Matrix Heatmaps for CCivilService Models

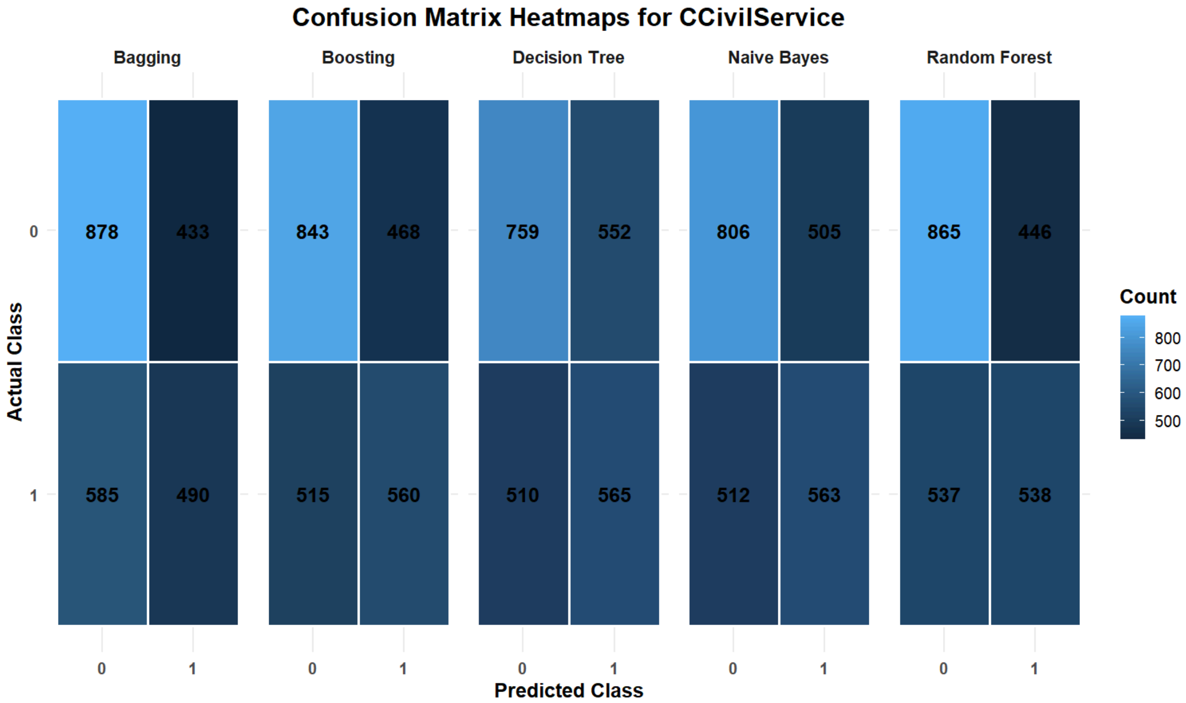


Figure 13: Confusion Matrix Heatmaps for CChurches Models

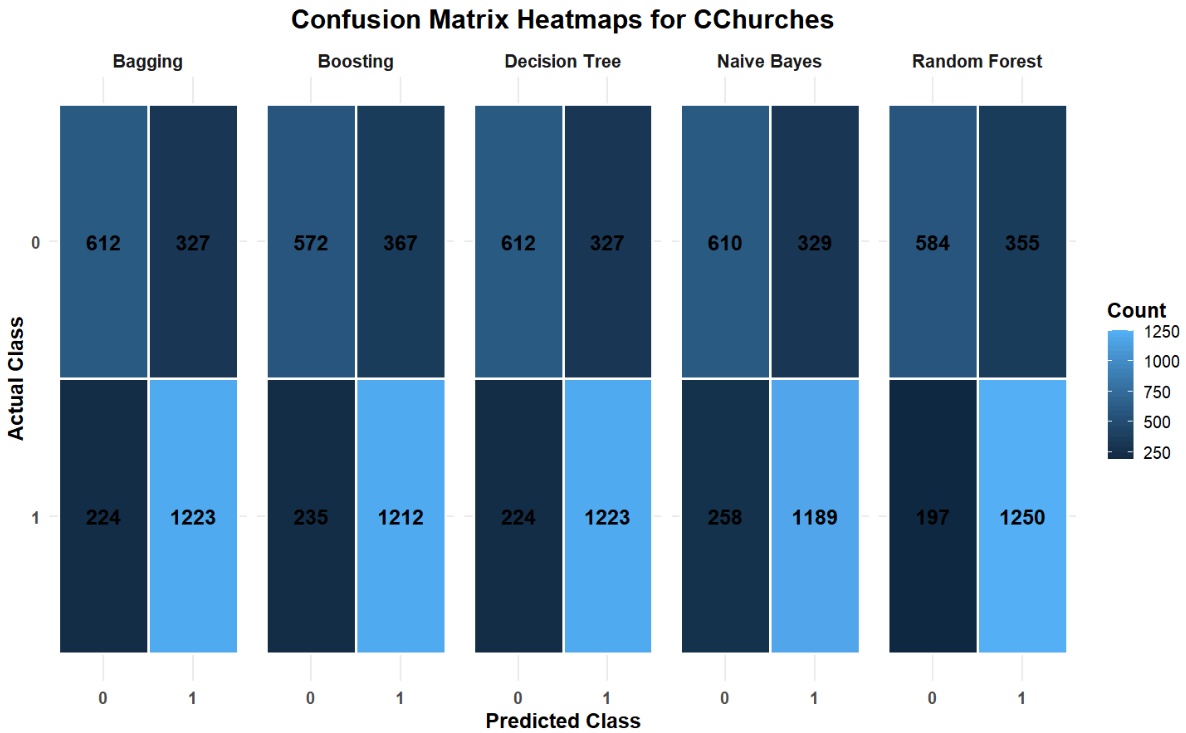


Figure 14: Confusion Matrix Heatmaps for CArmedForces Models

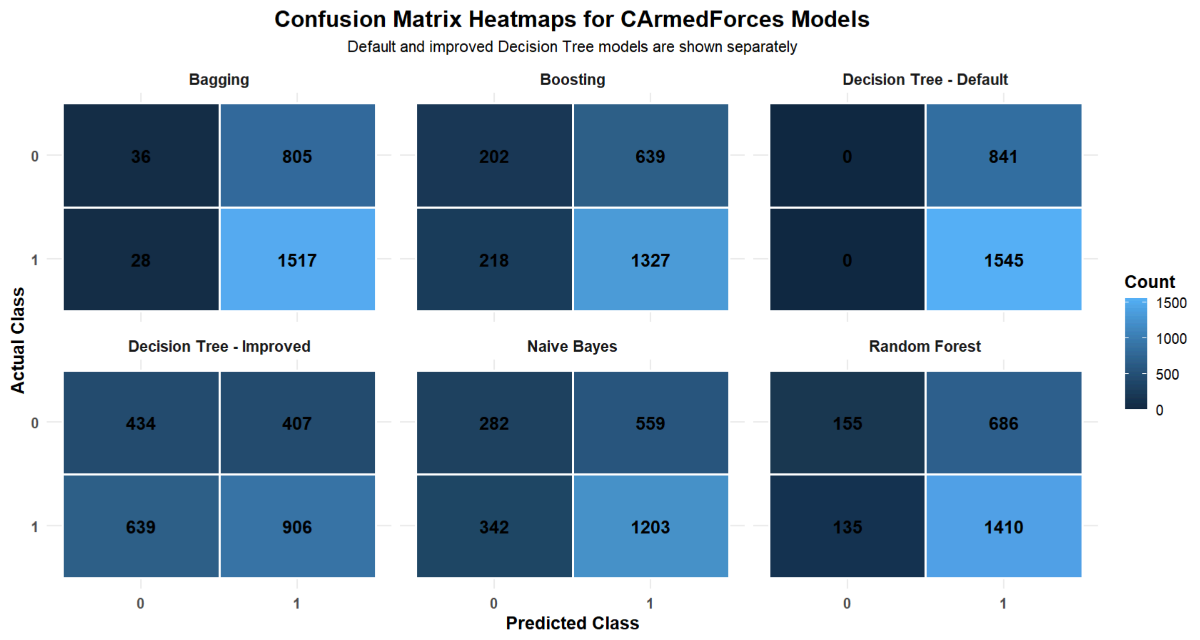


Table 5: Performance Metrics by Class Variable and Model

a) CCivilService

Model	Accuracy	Precision	Recall	F1-Score
Decision Tree	0.5549	0.5058	0.5256	0.5155
Naïve Bayes	0.5738	0.5272	0.5237	0.5254
Bagging	0.5733	0.5309	0.4558	0.4905
Boosting	0.588	0.5447	0.5209	0.5326
Random Forest	0.588	0.5467	0.5005	0.5226

b) CChurches

Model	Accuracy	Precision	Recall	F1-Score
Decision Tree	0.7691	0.789	0.8452	0.8161
Naïve Bayes	0.754	0.7833	0.8217	0.802
Bagging	0.7691	0.789	0.8452	0.8161
Boosting	0.7477	0.7676	0.8376	0.8011
Random Forest	0.7687	0.7788	0.8639	0.8191

c) CArmedForces

Model	Accuracy	Precision	Recall	F1-Score
Decision Tree	0.5616	0.69	0.5864	0.634
Naïve Bayes	0.6224	0.6827	0.7786	0.7275
Bagging	0.6509	0.6533	0.9819	0.7846
Boosting	0.6408	0.675	0.8589	0.7559
Random Forest	0.6559	0.6727	0.9126	0.7745

After fitting the classification models from Question 4, each model was evaluated on the test data for the three class variables: CCivilService, CChurches and CArmedForces. The confusion matrix heatmaps are shown in Figures 12 to 14, while the calculated accuracy, precision, recall and F1-score values are summarised in Table 5. Since the class distributions are not perfectly balanced, especially for CChurches and CArmedForces, F1-score and recall were prioritised over accuracy because they better show whether the model correctly identifies High confidence cases rather than simply favouring the majority class.

Table 6: Best and Worst Model Comparison Across Class Variables

Table 6 compares the strongest and weakest models for each class variable using the metrics in Table 5 and confusion matrix heatmaps in Figures 12 to 14.

Class Variable	**Best Model**	**Worst Model**
CCivilService	• All models performed similarly with accuracy ranging from 0.5549 to 0.5880 (Table 5(a)). • Boosting was the best model, achieving the highest F1-score of 0.5326 and joint-highest accuracy of 0.5880 (Table 5(a)). • Boosting generated 468 false positives and 515 missed actual positives, reflecting the limited separability of this class variable (Figure 12). • Naïve Bayes was second-best with an F1-score of 0.5254 and accuracy of	• Bagging was the worst model with the lowest F1-score of 0.4905, despite reasonable accuracy of 0.5733 (Table 5(a)). • It produced the most missed actual positives among all models for this class variable, with 585 false negatives (Figure 12).

	0.5738 (Table 5(a)).	
CChurches	• Performance was stronger across all models (Table 5(b)). • Random Forest was the best, achieving the highest F1-score of 0.8191 and recall of 0.8639, making it the most effective at identifying actual High confidence cases while keeping false positives relatively low (Table 5(b)). • Random Forest correctly classified 1250 High confidence cases and 584 Low confidence cases (Figure 13). • Decision Tree and Bagging were joint second-best with F1-scores of 0.8161, each correctly classifying 1223 High confidence and 612 Low confidence cases (Table 5(b) and Figure 13).	• Boosting was the worst model with the lowest F1-score of 0.8011 and accuracy of 0.7477 (Table 5(b)). • It missed more actual High confidence cases and generated more false positives than the stronger models (Figure 13).
CArmedForces	• Bagging achieved the best F1-score of 0.7846, driven by a very high recall of 0.9819, meaning it missed very few actual High confidence cases (Table 5(c)). • However, it correctly identified only 36 Low confidence cases against 805 false positives, showing heavy bias toward class 1 (Figure 14). • Random Forest was second-best with an F1-score of 0.7745 and the highest accuracy of 0.6559, correctly identifying 155 Low confidence cases while still capturing 1410 High confidence cases (Table 5(c) and Figure 14). This offered a more balanced trade-off between false positives and missed actual positives.	• Decision Tree was the worst model with the lowest F1-score of 0.6340 and accuracy of 0.5616 (Table 5(c)). • The default tree predicted only class 1, producing 841 false positives and missing all actual Low confidence cases entirely (Figure 14). • The improved Decision Tree produced a more balanced result but remained the weakest model overall (Figure 14 and Table 5(c)).

Overall, ensemble models generally outperformed single classifiers across all three class variables. Boosting was most appropriate for CCivilService, Random Forest for CChurches, and while Bagging achieved the highest F1-score for CArmedForces, Random Forest provided a more balanced and reliable performance given the class imbalance present in the data.

6. ROC Curves and AUC

Table 7: AUC Scores for Classification Models Across Three Class Variables

Class Variable	Decision Tree	Naïve Bayes	Bagging	Boosting	Random Forest
CCivilService	0.5523	0.6057	0.5889	0.622	0.6103
CChurches	0.7914	0.7925	0.7552	0.7909	0.8128
CArmedForces	0.5731	0.6106	0.5978	0.621	0.6211

Figure 15: ROC Curves for Predicting Confidence in CCivilService

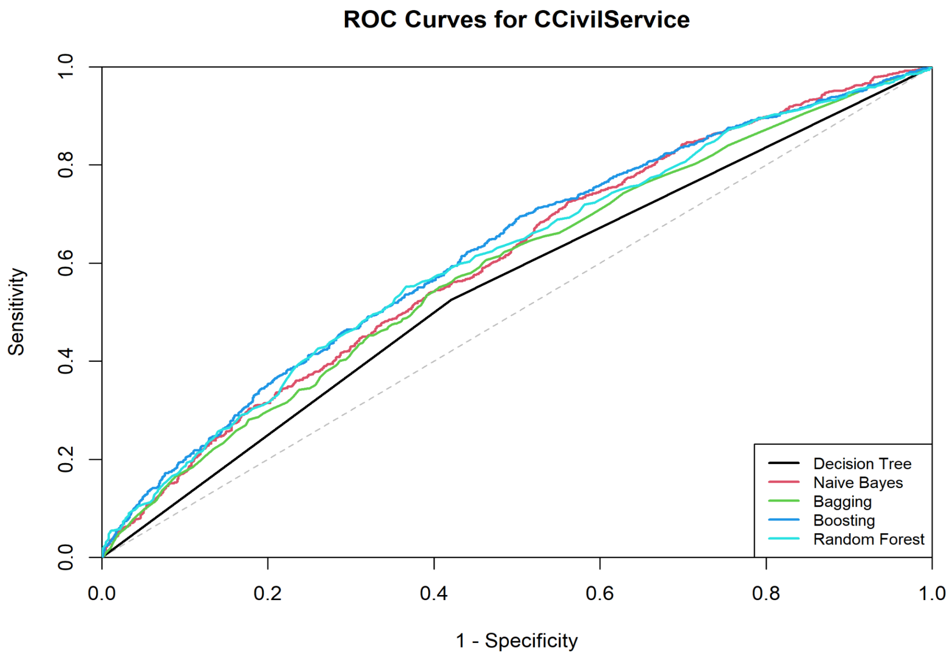


Figure 16: ROC Curves for Predicting Confidence in CChurches

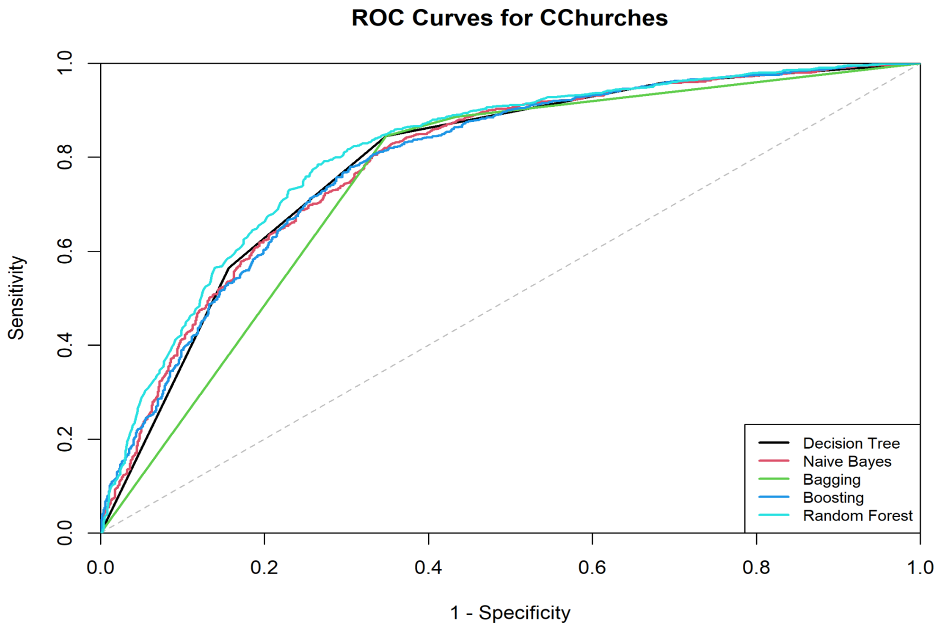
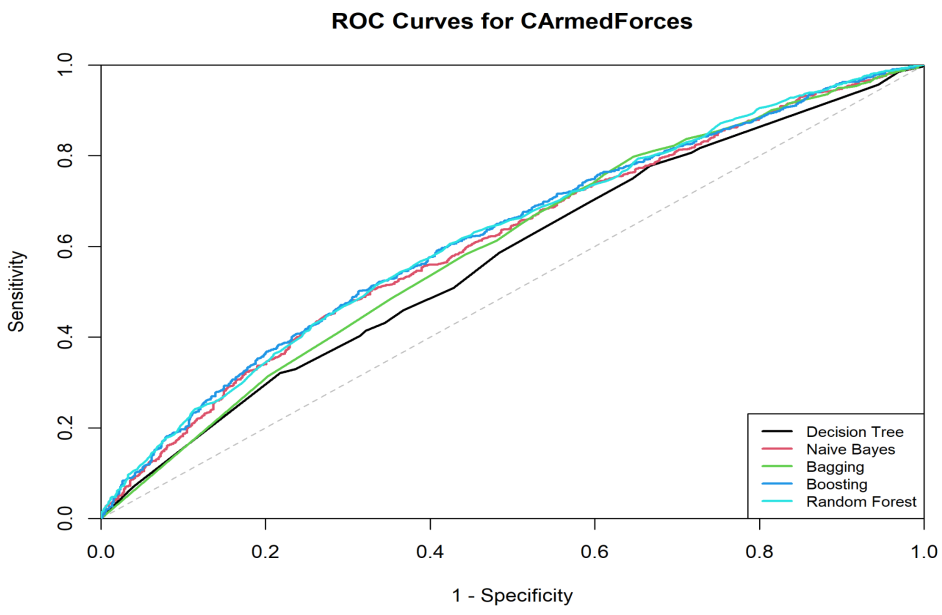


Figure 17: ROC Curves for Predicting Confidence in CArmedForces



Prediction probabilities were obtained from each classifier using the test dataset and used to construct ROC curves for the three class variables. The ROC curves are shown in Figures 15, 16, and 17, while the AUC values are summarised in Table 7. AUC measures how well a model separates the High confidence class from the Low confidence class across different classification thresholds, with values closer to 1 indicating stronger discriminatory ability and values closer to 0.5 indicating near-random separation.

Table 8: ROC and AUC Comparison Across Class Variables

Table 8 compares the AUC performance of each model for CCivilService, CChurches and CArmedForces, and summarises the key ROC curve patterns shown in Figures 15 to 17.

Class Variable	Best Model	Other Models	Worst Model	ROC Curve Observation
CCivilService	Boosting (AUC = 0.6220) (Table 7)	Random Forest (0.6103), Naïve Bayes (0.6057), Bagging (0.5889)	Decision Tree (AUC = 0.5523)	All curves sit only slightly above the diagonal baseline (Figure 15), indicating that no model could reliably distinguish between High and Low confidence cases. Boosting and Random Forest curves are furthest from the diagonal, suggesting marginally better discrimination, while Decision Tree runs closest to it, meaning it performed near randomly for this class variable.
CChurches	Random Forest (AUC = 0.8128) (Table 7)	Naïve Bayes (0.7925), Decision Tree (0.7914), Boosting (0.7909)	Bagging (AUC = 0.7552)	Curves are clearly furthest above the diagonal across all three class variables (Figure 16), indicating the strongest discriminatory ability overall. Random Forest sits furthest from the diagonal, confirming it best separates High and Low confidence cases. Bagging is noticeably more separated from the other four curves and closest to the diagonal, suggesting it struggled more with class separation despite reasonable F1-score driven by high recall.
CArmedForces	Random Forest and Boosting virtually equal (AUC = 0.6211 and 0.6210) (Table 7)	Naïve Bayes (0.6106), Bagging (0.5978)	Decision Tree (AUC = 0.5731)	Curves sit moderately above the diagonal (Figure 17), reflecting better-than-random but limited separation. Random Forest and Boosting sit furthest from the diagonal, confirming stronger discriminatory ability. Decision Tree is most visibly separated from the other curves and closest to the diagonal, consistent with its lowest AUC and weakest overall performance for this class variable.

Overall, CChurches was the easiest class variable to classify across all models, while CCivilService was the most difficult with AUC values closest to 0.5. Random Forest was the strongest overall model, achieving the highest AUC for both CChurches and CArmedForces and second-highest for CCivilService. Decision Tree was the weakest overall, producing the lowest AUC for CCivilService and CArmedForces.

7. Model Comparison Based on both Metrics and Auc

Table 9: Overall Model Comparison Across the Three Class Variables

Model	Average Accuracy	Average Precision	Average Recall	Average F1-Score	Average AUC
Random Forest	0.671	0.666	0.759	0.705	0.681
Boosting	0.659	0.662	0.739	0.696	0.678
Bagging	0.664	0.658	0.761	0.697	0.647
Naïve Bayes	0.65	0.664	0.708	0.685	0.67
Decision Tree	0.628	0.662	0.652	0.655	0.639

Table 9 compares the five classifiers using average results from Questions 5 and 6. Since the class variables were moderately imbalanced, recall and F1-score were prioritised over accuracy, with AUC used as supporting evidence. A model can achieve reasonable accuracy simply by favouring the majority class, so accuracy alone was treated cautiously.

Table 10: Comparative Analysis of Model Performance Using Metrics and AUC

Table 10 compares the best and worst models for each class variable using F1-score, recall, precision and AUC. It summarises which models performed strongest or weakest for CCivilService, CChurches and CArmedForces.

Class Variable	Best Model	Worst Model
CCivilService	• Boosting achieved the highest F1-score of 0.5326 and AUC of 0.6220 (Table 5(a) and Table 7). • Both recall and precision remained moderate, meaning the model still missed a	• Bagging was the worst model with F1 = 0.4905 and the lowest recall of 0.4558 (Table 5(a)). • Despite reasonable accuracy,

	notable number of actual High confidence cases while making incorrect High confidence predictions. • This reflects the weaker predictor separation observed during exploratory analysis, where CCivilService showed the least class-based distributional differences.	it missed too many actual High confidence cases, producing the most missed actual positives among all models for this class variable.
CChurches	• Random Forest achieved the highest F1-score of 0.8191, recall of 0.8639, and AUC of 0.8128 (Tables 5(b) and 7). • Strongest performance across all models, likely because religion-related predictors, particularly ILReligion, provided clearer class separation, reducing classification uncertainty.	• Boosting was the worst model with the lowest F1-score of 0.8011 (Table 5(b)). • The gap between models was small, suggesting all classifiers handled CChurches relatively well.
CArmedForces	• Bagging achieved the highest F1-score of 0.7846 and recall of 0.9819 (Table 5(c)), showing very strong detection of actual High confidence cases. • However, precision of only 0.6533 indicates it was over-predicting the majority class and producing many false positives for Low confidence cases. • Random Forest was second-best with F1 = 0.7745 and highest AUC = 0.6211 (Table 5(c) and Table 7) , offering a more balanced trade-off between correctly identifying High confidence cases and avoiding false positives.	• Decision Tree was the worst model with the lowest F1-score of 0.6340 (Table 5(c)). • The default tree initially predicted only class 1, confirming poor separation between the two confidence groups and making it the least suitable model for this imbalanced class variable.

There was no single best classifier across all three class variables, as the best model varied depending on class imbalance and predictor relationships for each outcome. These differences reflect the bias-variance trade-off. Decision Tree has high variance and is sensitive to weak predictor separation. Naïve Bayes assumes conditional independence, which was unlikely to hold given the correlated predictors identified earlier. Bagging reduces variance by averaging trees but still favours the dominant class under imbalance. Boosting reduces bias by focusing on difficult cases, which helps for CCivilService. Random Forest reduces variance further through bootstrap sampling and random predictor selection, making it the most robust overall.

Table 11: Overall Best and Worst Classifier Comparison

Table 11 compares the best and worst classifiers overall using average of F1-score, AUC and recall across the three class variables.

Category	Classifier	Avg F1	Avg AUC	Avg Recall	Explanation
Best Overall	Random Forest	0.705	0.681	0.759	Combining predictions from many trees trained on different data subsets with random predictor selection at each split made it more robust to correlated predictors and class imbalance across all three class variables than any single-tree approach.
Worst Ensemble	Bagging Specially for CCivilService	0.697	0.647	0.761	Reduces variance by averaging bootstrap trees but does not address class imbalance or correct misclassified cases, making it susceptible to majority class bias. This is evident from its very high recall but low precision for CArmedForces, where it produced a large number of false positives for Low confidence cases rather than learning meaningful decision boundaries. More specifically, for CCivilService, Bagging was the worst-performing ensemble model because it had the lowest F1-score (0.4905) and the lowest AUC (0.5889) among the ensemble classifiers for that class variable.
Worst Overall	Decision Tree	0.655	0.639	0.652	Structure depends on a small number of splits, making it susceptible to high variance and noise. The default tree for CArmedForces predicted only the majority class, reflecting its inability to handle class imbalance without additional adjustments, unlike ensemble methods that aggregate multiple models for more stable and balanced predictions.

8. Attribute Importance Across Ensemble Models

Figure 18: Attribute Importance for Predicting Confidence CCivilService Across Ensemble Models

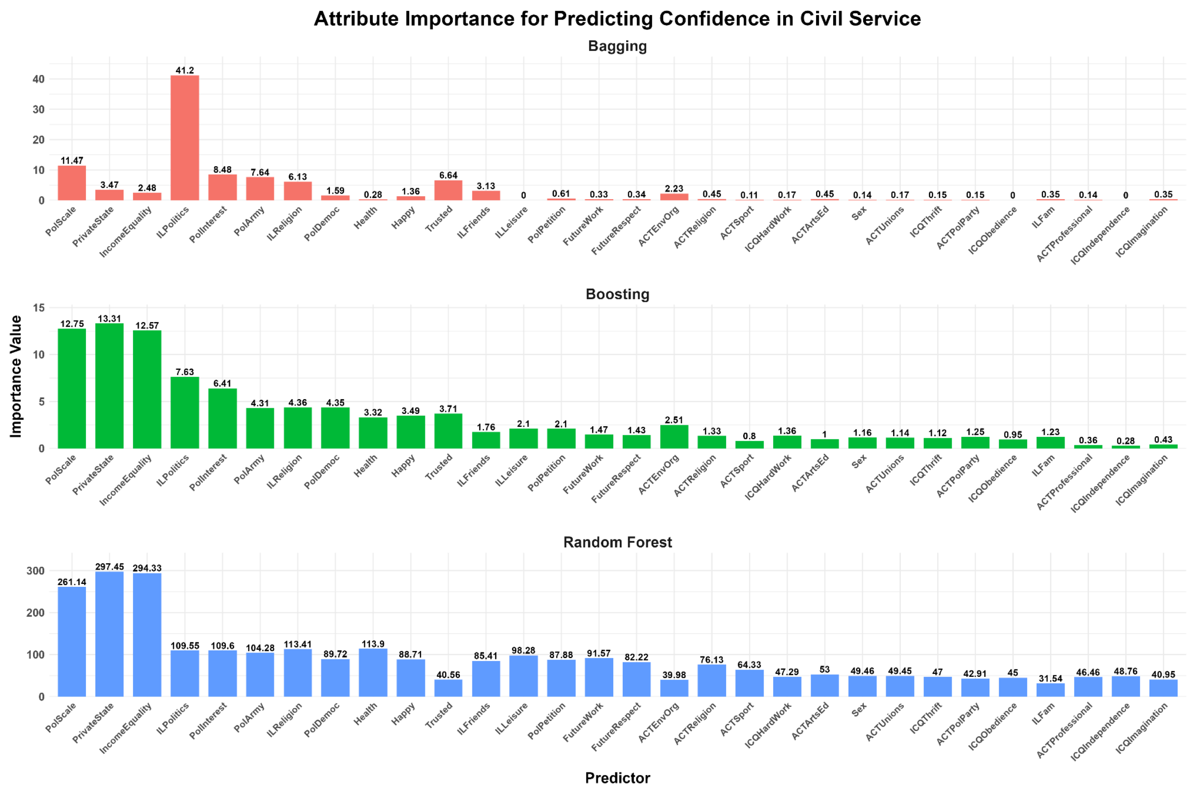


Figure 19: Attribute Importance for Predicting Confidence in CChurches Across Ensemble Models

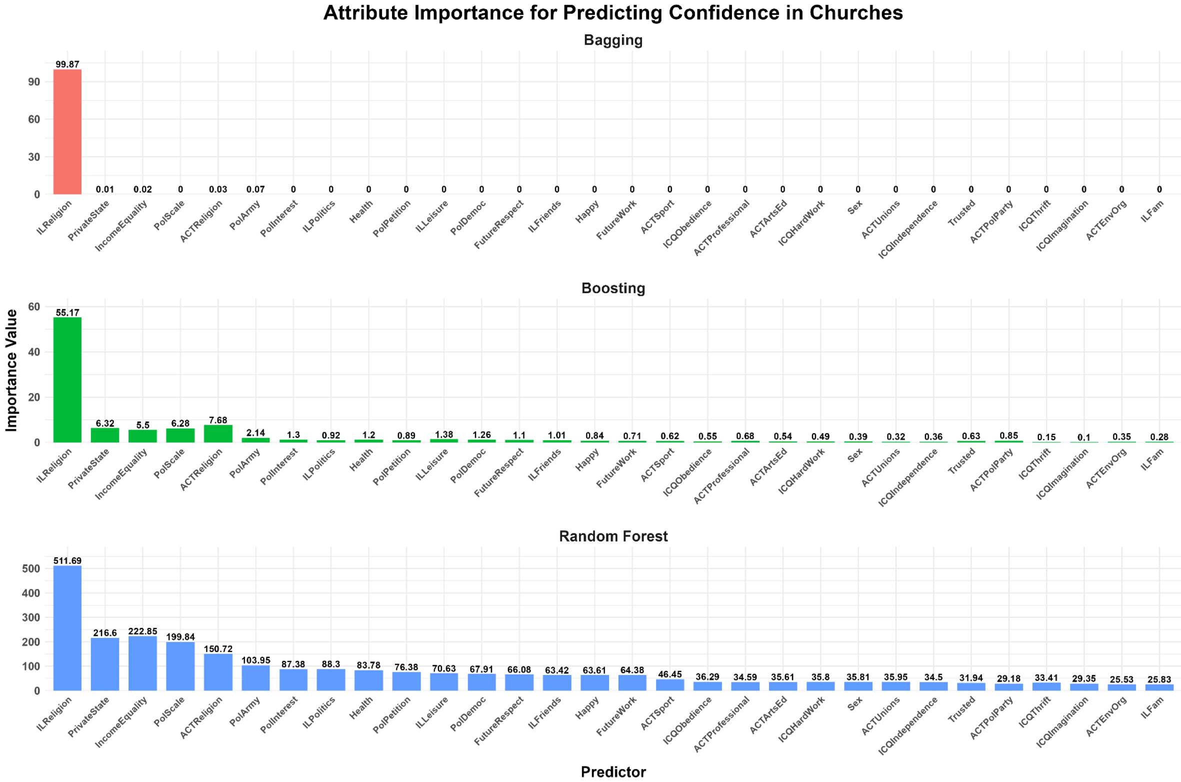
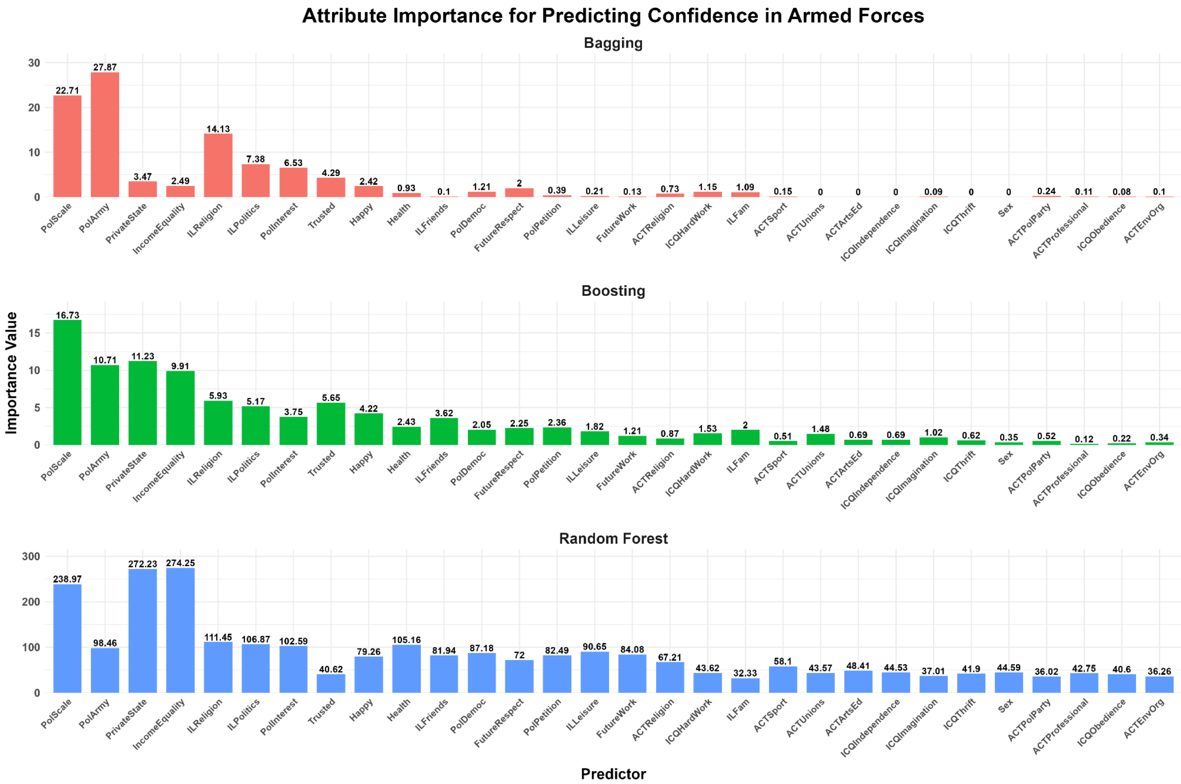


Figure 20: Attribute Importance for Predicting Confidence in Armed Forces Across Ensemble Models



Figures 18, 19, and 20 show the attribute importance values across the three ensemble models: Bagging, Boosting, and Random Forest. The figures are interpreted separately for each class variable. Importance values should be compared within each model rather than across models, as each method uses a different scale. Random Forest produces larger numerical values, so its predictors are interpreted relative to the strongest Random Forest predictors only.

Table 12: Most Important Attributes for CCivilService Across Ensemble Models

Ensemble Model	Most Important Attributes
Bagging	PolScale, PrivateState, IncomeEquality, ILPolitics, PolInterest, PolArmy, ILReligion, Trusted, ILFriends, ACTEnvOrg
Boosting	PolScale, PrivateState, IncomeEquality, ILPolitics, PolInterest, PolArmy, ILReligion, PolDemoc, Health, Happy, Trusted, ACTEnvOrg
Random Forest	PolScale, PrivateState, IncomeEquality, ILPolitics, PolInterest, PolArmy, ILReligion, Health, ILFriends, ILLeisure, PolPetition, FutureWork, FutureRespect,PolDemoc,Happy.

Table 13: Most Important Attributes for CChurches Across Ensemble Models

Ensemble Model	Most Important Attributes
Bagging	ILReligion
Boosting	ILReligion, ACTReligion, IncomeEquality, PrivateState, PolScale,PolArmy
Random Forest	ILReligion, PrivateState, IncomeEquality, PolScale, ACTReligion, PolArmy, PolInterest, ILPolitics, Health.

Table 14: Most Important Attributes for CArmedForces Across Ensemble Models

Ensemble Model	Most Important Attributes
Bagging	PolScale, PolArmy, PrivateState, IncomeEquality, ILReligion, ILPolitics, PolInterest, Trusted, Happy, FutureRespect.
Boosting	PolScale, PolArmy, PrivateState, IncomeEquality, ILReligion, ILPolitics, PolInterest, Trusted, Happy, Health, ILFriends, PolDemoc, FutureRespect, PolPetition, ILFam.
Random Forest	PolScale, PolArmy, PrivateState, IncomeEquality, ILReligion, ILPolitics, PolInterest, Health, ILFriends, PolDemoc, PolPetition, ILLeisure, FutureWork.

Table 15: Most Important Attribute Patterns Across Class Variables

Table 15 summarises the most important predictors identified from the ensemble model importance results.

Class Variable	Most Important Attributes
CCivilService (Figure 18, Table 12)	• ILPolitics, PolScale, PolInterest, PolArmy, PrivateState, IncomeEquality, Trusted, ILReligion, and Health appeared consistently across all three models. • In Bagging, ILPolitics = 41.20 was the highest, with political, trust, and religious variables dominating. • In Boosting, PrivateState = 13.31 was the highest, with state-related, economic, and political variables dominating. • In Random Forest, PrivateState = 297.45 was the highest, with state-related, economic, political, and health variables dominating. • Overall, confidence in civil service is mainly linked to political position, state responsibility views, and income equality attitudes, suggesting respondents with stronger political views and opinions about government responsibility tend to have more defined confidence levels in the civil service.
CChurches (Figure 17, Table 13)	• ILReligion, ACTReligion, IncomeEquality, PrivateState, PolScale, and PolArmy appeared consistently across all three models. • In Bagging, ILReligion = 99.87 was the highest by a large margin, with religious variables almost entirely dominating and all other predictors close to zero. • In Boosting, ILReligion = 55.17 was the highest, with religious, state-related, and economic variables dominating. • In Random Forest, ILReligion = 511.69 was the highest, with religious, state-related, and economic variables dominating. • This is consistent with the class-based separation observed in Figure 3b, confirming that religious importance was the dominant driver of confidence in churches.
CArmedForces (Figure 20, Table 14)	• PolArmy, PolScale, ILReligion, ILPolitics, PolInterest, PrivateState, IncomeEquality, Trusted, and Health appeared consistently across all three models.

	• In Bagging, PolArmy = 27.87 was the highest, with army-related, political, and religious variables dominating. • In Boosting, PolScale = 16.73 was the highest, with political, state-related, and economic variables dominating. • In Random Forest, IncomeEquality = 274.25 was the highest, with state-related, economic, political, and religious variables dominating. • Overall, confidence in the armed forces is strongly connected to army attitudes, political orientation, and state responsibility views, suggesting respondents with stronger support for authority tend to have more defined confidence levels in the armed forces.

Table 16: Least Important Attributes for CCivilService Across Ensemble Models

Ensemble Model	**Least Important Attributes**
Bagging	ILLeisure, ICQObedience, ICQIndependence, ACTSport, Sex, ICQHardWork, ACTUnions, ICQThrift, ACTPolParty, ACTProfessional
Boosting	ICQIndependence, ACTProfessional, ICQImagination, ACTSport, ICQObedience
Random Forest	ILFam, ACTEnvOrg, Trusted, ICQImagination, ACTPolParty, ICQObedience

Table 17: Least Important Attributes for CChurches Across Ensemble Models

Ensemble Model	**Least Important Attributes**
Bagging	PolInterest, ILPolitics, Health, PolPetition, ILLeisure, PolDemoc, FutureRespect, ILFriends, Happy, FutureWork, ACTSport, ICQObedience, ACTProfessional, ACTArtsEd, ICQHardWork, Sex, ACTUnions, ICQIndependence, Trusted, ACTPolParty, ICQThrift, ICQImagination, ACTEnvOrg, ILFam
Boosting	ICQImagination, ICQThrift, ILFam, ACTEnvOrg, ACTUnions, Sex, ICQIndependence
Random Forest	ILFam, ACTEnvOrg, ACTPolParty, ICQImagination, ICQThrift

Table 18: Least Important Attributes for CArmedForces Across Ensemble Models

Ensemble Model	**Least Important Attributes**
Bagging	ACTUnions, ACTArtsEd, ICQIndependence, ICQThrift, Sex
Boosting	ICQObedience, ACTEnvOrg, ACTProfessional, Sex, ACTSport,

	ACTPolParty
Random Forest	ICQImagination, ICQThrift, ACTPolParty, ACTEnvOrg, ILFam, ICQObedience, Trusted

Table 19: Least Important Attribute Patterns Across Class Variables

Table 19 summarises the least important predictors across the ensemble models for the three class variables

Class Variable	**Least Important Attributes**
CCivilService (Table 16)	• ILLeisure, ICQObedience, ICQIndependence, ACTSport, ACTUnions, and ICQThrift appeared consistently with low importance across all three models. • These are mainly leisure, activity, and child quality variables, suggesting they do not meaningfully distinguish between Low and High confidence in civil service and could be omitted without substantially affecting performance.
CChurches (Table 17)	• ILFam, ACTEnvOrg, ACTPolParty, ICQImagination, and ICQThrift appeared with low or near-zero importance across all three models. • In Bagging, many variables had very low or zero importance because ILReligion dominated the model entirely, suggesting these activity, family, and child quality variables add little extra predictive value once religious importance is already included.
CArmedForces (Table 18)	• ACTUnions, ACTArtsEd, ICQIndependence, ICQThrift, ACTEnvOrg, ACTProfessional, ICQImagination, and ILFam appeared with low or near-zero importance across all three models. • These are mainly activity, child quality, and family variables, showing no consistent predictive signal across Bagging, Boosting, or Random Forest and contributing little to predicting confidence in the armed forces.

Overall Important Attributes

PolScale, PrivateState, IncomeEquality, ILPolitics, PolInterest, and ILReligion appeared consistently important across all three class variables, suggesting that institutional confidence is broadly driven by political attitudes, state and economic views, and religious importance. The dominant predictor varied by class variable. ILReligion dominated CChurches, reflecting the strong and intuitive link between religious importance and confidence in religious institutions. PolArmy was the highest predictor for CArmedForces in Bagging, reflecting the direct relationship between attitudes toward the military and confidence in armed forces. CCivilService was influenced by a wider mix of political, state-related, and economic variables rather than one single dominant predictor, which is consistent with the weaker class separation observed in Figure 3c.

Attributes That Could Be Omitted

Attributes repeatedly appearing with low or near-zero importance across Tables 16, 17, and 18 are strong candidates for omission. These include child quality variables such as ICQIndependence, ICQImagination, ICQThrift, and ICQObedience, and activity variables such as ACTUnions, ACTEnvOrg, ACTProfessional, ACTPolParty, and ACTSport. This is consistent with the correlation matrix in Figure 4, where ACT variables showed inter-predictor correlations of up to 0.96, confirming strong redundancy among activity variables. The child quality variables also showed limited variation in the overall predictor boxplots in Figure 1, further supporting their removal. Omitting these variables could simplify the models and reduce noise while retaining the stronger political, religious, and state-related predictors that consistently drove predictions across all three class variables.

9. Explaining Differences in Model Performance

The differences in model performance can be explained by each model's properties and its ability to handle non-linearity, predictor interactions, noise, and correlated variables. Figure 4 shows strong correlations among several predictors, with some pairs exceeding r = 0.90, meaning some predictors contain overlapping information. Models that can handle correlated and noisy predictors are therefore expected to perform better.

Table 20: Strengths and Weaknesses of Classification Models

Table 20 briefly compares each classifier’s strengths, weaknesses and performance. It links model behaviour to the dataset issues, including correlated predictors, class imbalance and differences in F1-score/recall.

Model	**Strengths & Weaknesses**	**Performance**
Decision Tree	**Strength:** Can capture non-linear relationships in the data through splitting, making it interpretable. **Weakness:** A single tree is unstable and prone to overfitting with noisy or correlated predictors like those in Figure 4, reducing its ability to generalise(Quinlan, 1986). Too simple for this dataset because CCivilService predictors were spread across several political and state-related variables rather than one dominant predictor.	Weakest overall; F1 = 0.5155 for CCivilService (Table 5(a)) and F1 = 0.6340 for CArmedForces (Table 5(c)).
Naïve Bayes	**Strength:** Works well when one predictor dominates, as with ILReligion for CChurches. **Weakness:** The conditional independence assumption is	F1 = 0.8020 for CChurches (Table 5(b)), but weaker for CArmedForces (F1 =

	violated here since Figure 4 shows correlated predictors in the data such as ACTPolParty and ACTEnvOrg = 0.96, limiting performance when multiple predictors interact (Rish, 2001).	0.7275, Table 5(c)) where prediction depended on a wider mix of predictors.
Bagging	**Strength:** Reduces variance by aggregating predictions across bootstrap samples, improving stability over a single tree. **Weakness:** When the predictor signal is weak and spread out, variance reduction alone is insufficient(Breiman, 1996). In this dataset, different class distributions affected balance; CArmedForces had stronger High confidence proportions than CCivilService.	Strongest for CArmedForces (F1 = 0.7846, recall = 0.9819, Table 5(c)), but weakest for CCivilService (F1 = 0.4905, Table 5(a)).
Boosting	**Strength:** Builds learners sequentially, focusing on harder-to-classify observations, which is beneficial when no single dominant predictor exists. **Weakness:** Can be sensitive to noisy or overlapping predictors present in Figure 4. May over-focus on noisy or overlapping survey patterns because misclassified cases receive more weight (Freund & Schapire, 1997).	Best for CCivilService (F1 = 0.5326, accuracy = 0.5880, Table 5(a)), but not consistently strongest across all class variables.
Random Forest	**Strength:** Random feature selection reduces the influence of correlated predictors shown in Figure 4, making it robust to noise and non-linearity. **Weakness:** Less interpretable than a single Decision Tree (Breiman, 2001).	Most consistently strong; best CChurches results (F1 = 0.8191, recall = 0.8639, Table 5(b)) and strong CArmedForces performance (F1 = 0.7745, Table 5(c)).

Overall, models capable of handling non-linearity, interactions, and correlated predictors, particularly Random Forest, Boosting, and Bagging, consistently outperformed simpler approaches.

10. Improving the Worst-Performing Ensemble Model

Table 21: Cross-Validation Comparison of Predictor Sets for Improved Bagging Model

Predictor Set	Mean CV Accuracy	Mean CV Precision	Mean CV Recall	Mean CV F1-score	Mean CV AUC
All Predictors	0.5685	0.5558	0.479	0.5135	0.5832

Selected Strongest Predictors	0.5696	0.5584	0.4619	0.5053	0.5843
Selected Strongest Predictors Without High Correlation	0.5724	0.562	0.4679	0.51	0.581

Figure 21: ROC Curve Comparison for Original and Improved Bagging Model for CCivilService

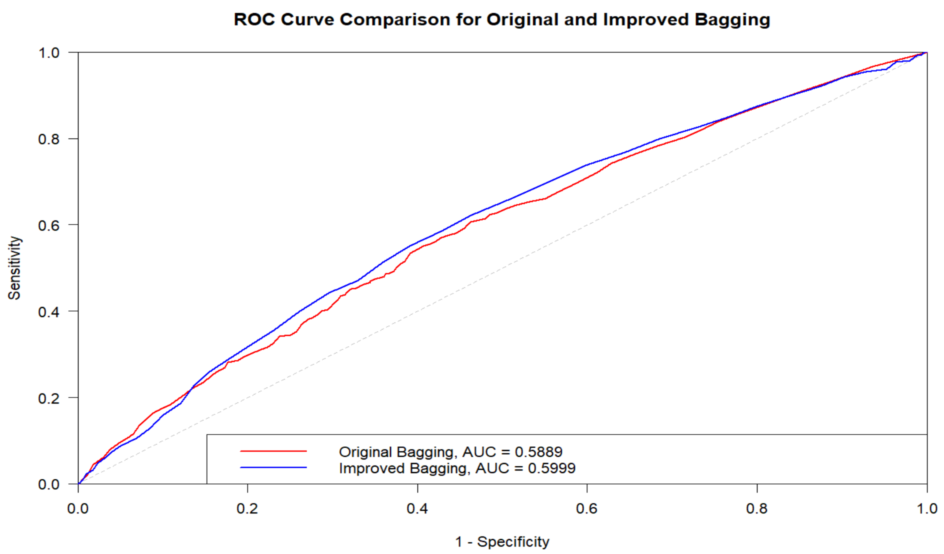


Table 22: Performance Comparison of Original and Improved Bagging Model for CCivilService

Model	Accuracy	Precision	Recall	F1-score	AUC
Original Bagging	0.5733	0.5309	0.4558	0.4905	0.5889
Improved Bagging	0.5109	0.476	0.8502	0.6104	0.5999

The model selected for improvement was the Bagging model for CCivilService, as it was the weakest ensemble model with the lowest F1-score of 0.4905 (Table 5(a)) and AUC of 0.5889 (Table 7), compared to Boosting (F1 = 0.5326, AUC = 0.6220) and Random Forest (F1 = 0.5226, AUC = 0.6103). F1-score and AUC were prioritised because F1-score balances precision and recall for the High confidence class, while AUC evaluates how well the model separates Low and High confidence cases across thresholds, making them more reliable than accuracy alone when classification errors and class balance are important.

The improved model was built in R using the ipred, rpart, pROC, and dplyr packages, with Country, Wave, and other class variables excluded to remain consistent with earlier models. Three predictor sets were tested: All Predictors, Selected Strongest Predictors from Question 8, and Selected Strongest Predictors Without High Correlation, where ACTEnvOrg was removed to reduce redundancy given the correlations shown in Figure 4. A 3-fold cross-validation tuned combinations of nbagg (50 or 100), cp (0.001 or 0.005), minsplit (10 or 20), and maxdepth (5 or 8), with models ranked primarily by mean CV F1-score. As shown in Table 21, All Predictors achieved the highest mean CV F1-score of 0.5135 and was selected for the final model. The selected predictor sets did not improve F1-score further, suggesting that removing predictors caused more information loss than noise reduction for this class variable, which relied on a broader mix of signals rather than any single dominant predictor.

The final model was trained using All Predictors, nbagg = 50, cp = 0.001, minsplit = 10, and maxdepth = 8. The classification threshold was then adjusted from 0.50 to 0.30 to address the original model's low recall of 0.4558, without retraining the model. Since CCivilService was the most balanced class variable but still had some imbalance (52.19% Low, 47.81% High, Table 1), F1-score was prioritised over accuracy as it better balances precision and recall.

As shown in Table 22, the improved model achieved an F1-score of 0.6104 and recall of 0.8502, meaning it correctly identified far more respondents who actually hold High confidence in the civil service. AUC also improved from 0.5889 to 0.5999, confirming that the tuned model produced better probability rankings overall, not only as a result of the threshold change. The trade-off was a decrease in accuracy from 0.5733 to 0.5109 and precision from 0.5309 to 0.4760, as lowering the threshold increased true positives but also increased false positives. This trade-off was acceptable as, given the class imbalance in CCivilService, F1-score is a more appropriate evaluation metric than accuracy.

Figure 21 supports this, with the blue improved model curve sitting slightly further from the diagonal than the red original curve, consistent with the higher AUC and confirming genuine improvement in probability rankings beyond the threshold adjustment alone.

11. ANN Classifier for Predicting Confidence in Civil Service

Figure 22: Number of Observations by Country for ANN Country Selection

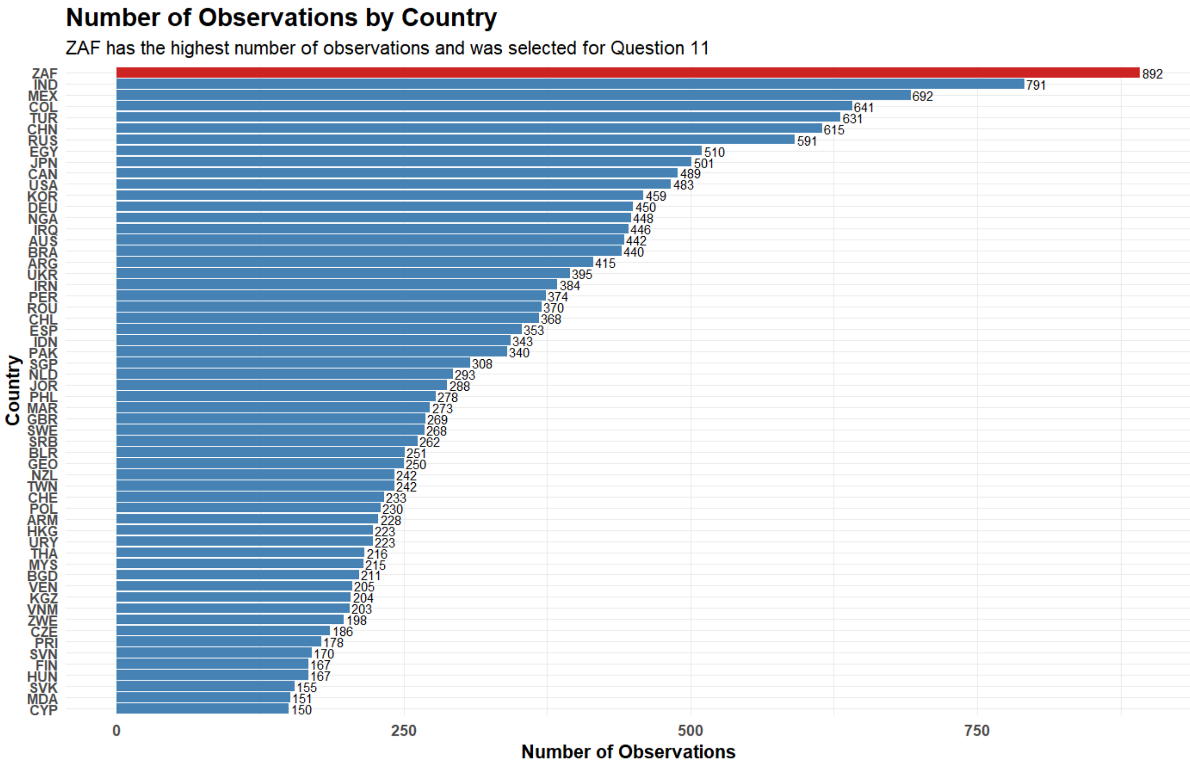


Figure 23: ANN Performance Comparison Between Selected Predictors and All Predictors

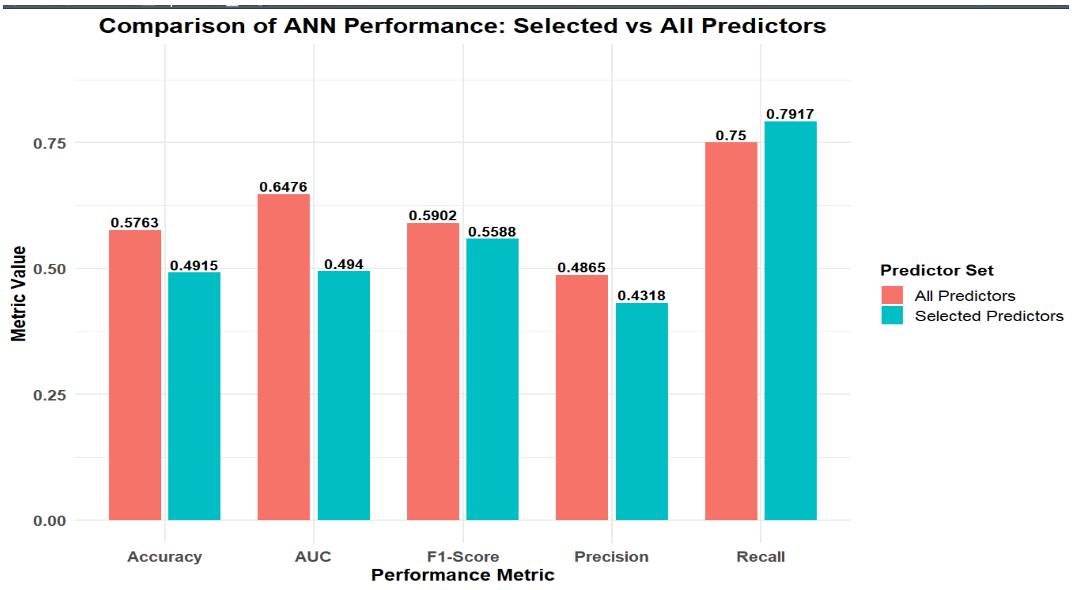
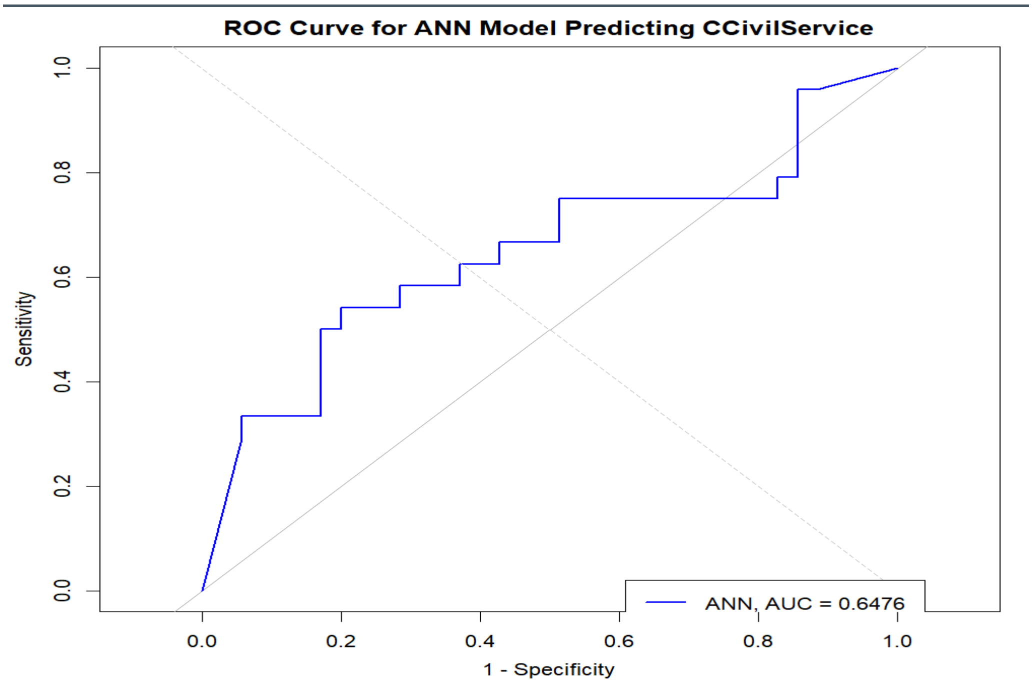


Table 23: Final ANN Model Performance for CCivilService in ZAF

Model	Accuracy	Precision	Recall	F1-score	AUC
ANN - All Predictors	0.5763	0.4865	0.75	0.5902	0.6476

Figure 24: ROC Curve for ANN Model Predicting CCivilService in ZAF



Country and Class Variable Selection:

ZAF (South Africa) was selected as it had the greatest number of observations after data cleaning, with 892 observations as shown in Figure 22. The target class variable was CCivilService, as it had already been the focus of the earlier ensemble models and the improved Bagging model in Question 10. The existing training and test split was retained and filtered to ZAF, resulting in 149 training observations and 59 test observations.

Data Preprocessing:

The ANN was implemented in R using the nnet package. A final complete-case check was applied to the selected ZAF records before fitting. Categorical predictors were converted into numeric dummy variables using model.matrix(), as ANN models require numeric inputs, resulting in 85 dummy-coded columns from 30 predictors. Predictors were then standardised using the training-set mean and standard deviation, with the same scaling applied to the test data, as ANN models are sensitive to variable scale. Class weights were applied during training to account for the mild imbalance in CCivilService (52.19% Low, 47.81% High, Table 1), giving higher weight to the minority class to prevent the model from being biased toward predicting Low confidence.

Predictor Selection:

Two predictor sets were tested. The selected-predictor ANN used variables identified from the earlier CCivilService attribute importance analysis shown in figure 18 and table 12: PolScale, PrivateState, IncomeEquality, ILPolitics, PolInterest, PolArmy, ILReligion, PolDemoc, Health, Trusted, ILFriends, ILLeisure, PolPetition and FutureWork. These were chosen because they consistently ranked as important across the Bagging, Boosting and Random Forest models for CCivilService. As shown in Figure 23, the all-predictor ANN outperformed the selected-predictor ANN on accuracy (0.5763 vs 0.4915), precision (0.4865 vs 0.4318), F1-score (0.5902 vs 0.5588) and AUC (0.6476 vs 0.4940), with only recall being higher for selected predictors (0.7917 vs 0.7500). The final model therefore used all valid predictors after excluding Country, Wave and the class variables.

Model Architecture and Tuning:

The ANN used one hidden layer with softmax output activation, which produces class probabilities suitable for binary classification. A tuning grid was tested across hidden layer sizes (3, 5, 7, 10, 12), decay values (0, 0.001, 0.01, 0.05, 0.10) and maxit values (500, 1000), with an internal 80:20 validation split from the training data used to select the best combination. The final best settings were size = 3, decay = 0, and maxit = 1000. A hidden layer of three neurons was appropriate given the small ZAF training sample of 149 observations, as larger networks risked overfitting. The classification threshold was tuned separately and set to 0.24 to improve recall and maximise F1-score.

Model Performance and Comparison:

As shown in Table 23, the final ANN achieved an accuracy of 0.5763, precision of 0.4865, recall of 0.7500, F1-score of 0.5902, and AUC of 0.6476. However, the lower precision of 0.4865 shows that the model also produced false positives, meaning some Low confidence respondents were incorrectly classified as High confidence. Table 24 compares the ANN against all earlier classifiers for CCivilService.

Table 24: Comparison of ANN with Earlier Classifiers for CCivilService

Model	Accuracy	Precision	Recall	F1-Score	AUC
Decision Tree	0.5549	0.5058	0.5256	0.5155	0.5523
Naïve Bayes	0.5738	0.5272	0.5237	0.5254	0.6057
Bagging	0.5733	0.5309	0.4558	0.4905	0.5889
Boosting	0.588	0.5447	0.5209	0.5326	0.622

Random Forest	0.588	0.5467	0.5005	0.5226	0.6103
ANN	0.5763	0.4865	0.75	0.5902	0.6476

The ANN achieved the highest F1-score and the highest AUC across all models, showing the strongest balance between precision and recall and the best overall discriminatory ability for CCivilService. Figure 24 supports this, with the ROC curve sitting above the diagonal reference line, though the curve is not very close to the top-left corner, indicating moderate rather than very strong discrimination.

Reasons for Performance:

The ANN likely outperformed the earlier classifiers because it can capture non-linear relationships and interactions between predictors without assuming independence, unlike Naïve Bayes, or relying on simple splits, unlike Decision Trees. The all-predictor approach may also have helped because weaker predictors still contributed useful signals when combined through the network. However, the ANN's precision was relatively low at 0.4865, as the lowered threshold and small training sample made the model more likely to over-predict High confidence cases. Overall, the ANN achieved the strongest F1-score and AUC for CCivilService but its moderate precision reflects a trade-off between identifying true High confidence respondents and avoiding false positives.

12. ANN Performance Comparison Across Waves

Figure 25: Number of ZAF Test Observations by Wave

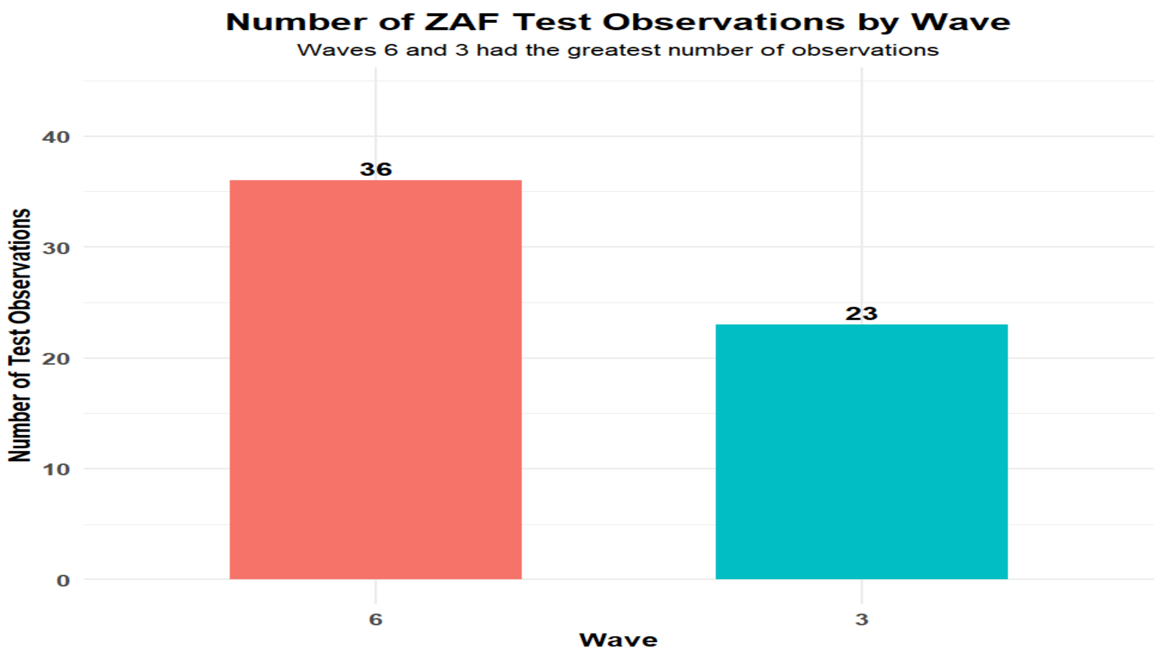


Figure 26: Number of ZAF Test Observations for the Two Selected Waves

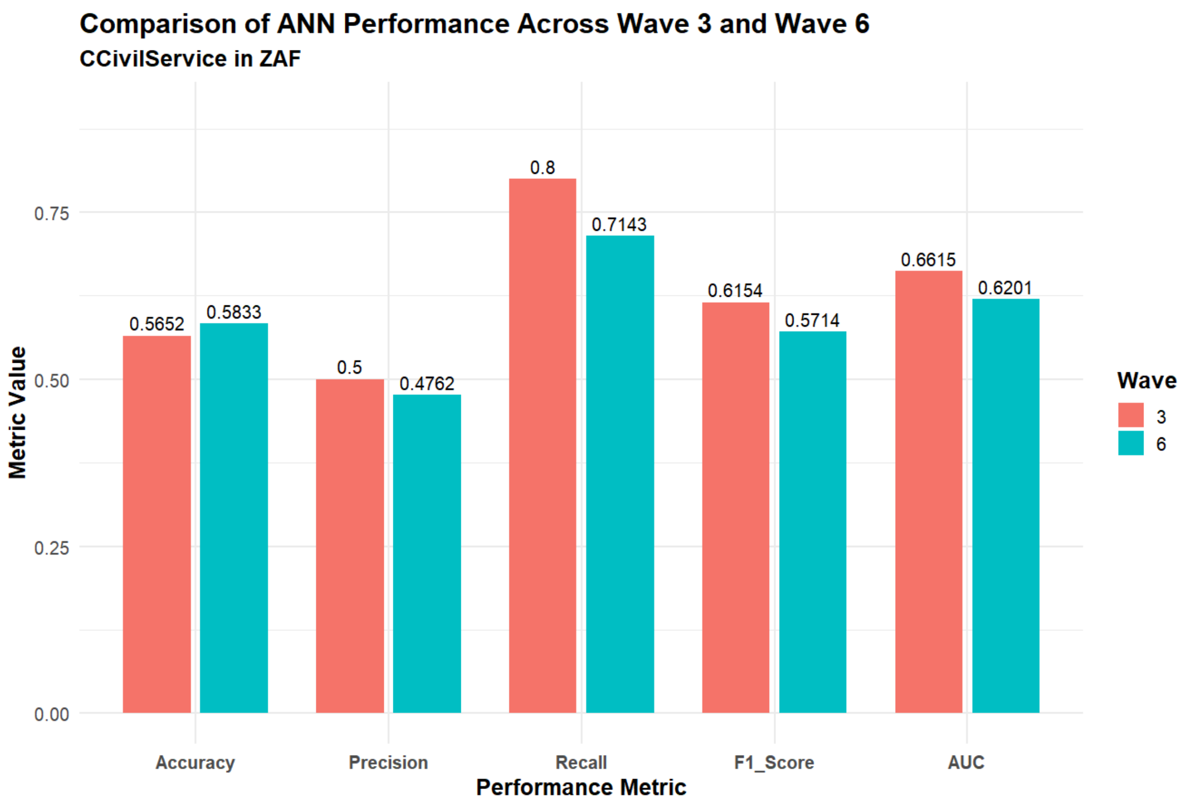
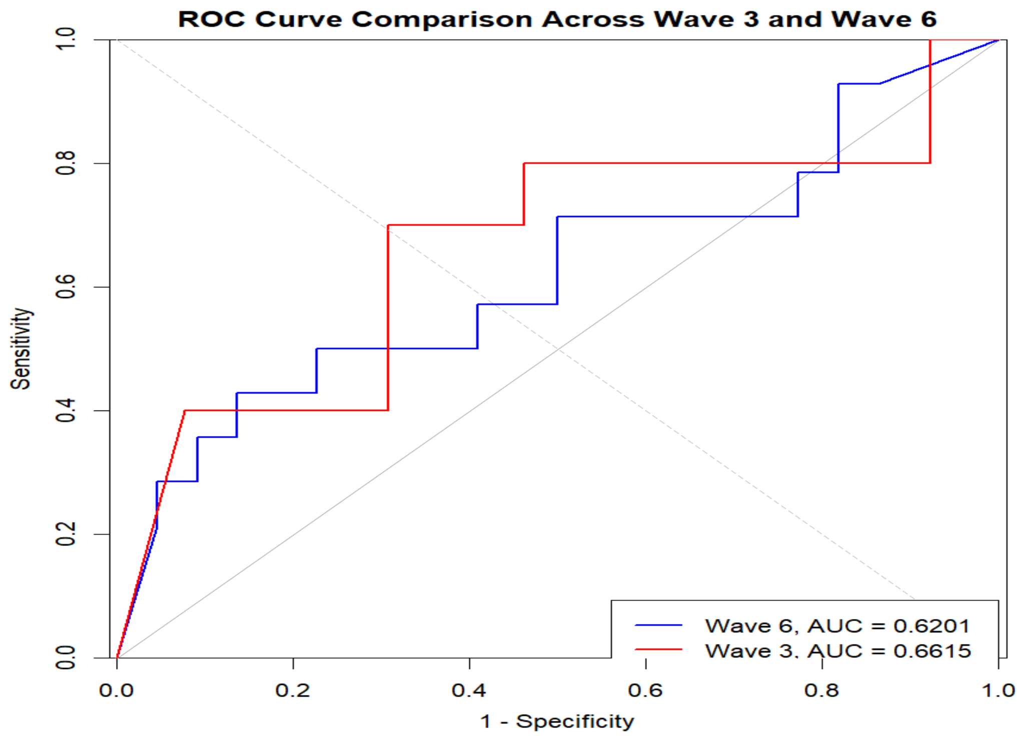


Table 25: ANN Performance Comparison Across Wave 3 and Wave 6

Wave	Number of Observations	Accuracy	Precision	Recall	F1-score	AUC
3	23	0.5652	0.5	0.8	0.6154	0.6615
6	36	0.5833	0.4762	0.7143	0.5714	0.6201

Figure 27: ROC Curve Comparison Across Wave 3 and Wave 6 for CCivilService in ZAF



The Artificial Neural Network model developed in Question 11 was used to classify CCivilService test instances for ZAF/South Africa. The same trained ANN model was applied separately to the ZAF test observations in each wave without retraining. Wave was only used to split the test data for comparison, not as an ANN predictor.

As shown in Figure 25, the ZAF test data contained observations from only two waves. Wave 6 had the greater number of test observations with 36 cases, while Wave 3 had 23 cases. These were therefore selected for the Q12 comparison.

The performance results are shown in Table 25 and Figure 26. The ANN performed better in Wave 3 than in Wave 6 on most classification metrics. Wave 3 achieved higher precision (0.5000 vs 0.4762), higher recall (0.8000 vs 0.7143) and a higher F1-score (0.6154 vs 0.5714). This indicates a more balanced classification result in Wave 3. Wave 6 had slightly higher accuracy (0.5833 vs 0.5652), but this difference is small, and accuracy alone is less informative because class distribution differs across waves.

The ROC curve comparison in Figure 27 supports this. Wave 3 achieved a higher AUC of 0.6615, compared with 0.6201 for Wave 6, meaning the ANN had better discriminatory ability in Wave 3. Both AUC values are above 0.5, so the model performed better than random classification in both waves, although the values remain moderate.

Overall, the ANN showed moderate stability over time, but its predictive performance was not identical across waves. The stronger Wave 3 F1-score and AUC suggest the model found Wave 3 cases easier to classify, while Wave 6 showed weaker separation between Low and High confidence cases. This may reflect data drift or temporal variation in the relationship between predictors and confidence in the civil service. Changes in political and institutional conditions in South Africa may have affected predictors such as PolScale, PolArmy and Trusted over time. The small wave-specific sample sizes also mean that a few changed predictions can noticeably affect the reported metrics, so the differences should be interpreted cautiously.

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